When Do Text Proxies Suffice? A Regime Analysis of Sparse, Dense, and Native-Visual Retrieval Under Proxy Noise

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Abstract—Multimodal Retrieval-Augmented Generation (MM-RAG) frequently indexes precomputed textual proxies (OCR transcripts, machine captions) in place of raw pixels. Across a standardized 1,626-item benchmark under 50/50 Dev/Test splits ($N = 306$ held-out probes, 813 test items), we investigate when text proxies suffice and when native-visual representations are indispensable. Contrary to common assumptions, character 3–5gram BM25 strictly dominates word BM25 across all verbatim optical OCR regimes (Clean: 0.9436 vs. 0.8371; Degraded: 0.8248 vs. 0.6972; Tesseract 5.5: 0.8410 vs. 0.7361; all $p_{\text{Holm}} < 10^{-4}$), reversing only under entity-substituting light paraphrasing (0.2427 vs. 0.4334). On ground-truth text, dense bi-encoder failure on verbatim spans (MRR 0.2100) is governed by query formulation mismatch; answerable questions restore BGE-Base MRR to 0.5197 (Hit@10 78.8%). Across 200 cluster-stratified splits, conformal cascade calibration under marginal loss rigorously bounds test miss risk below 0.05 (mean test risk $0.0491 \leq 0.05$) at 78.4% escalation savings. On MS-COCO ($N = 1{,}000$), native visual retrieval (SigLIP Base Patch16 MRR = 0.6071) decisively outperforms caption proxies (0.3728, $p < 10^{-4}$). In multi-hop reasoning, independent pseudo-relevance feedback matches generative LLM bridging (Hit@10 0.640 vs. 0.640, exact McNemar $p = 1.000$) with zero API cost.

Index Terms—Multimodal RAG, Text Proxies, Proxy Noise, Dense-Sparse Regime Analysis, Cascade Ranking, Conformal Risk Control, Multi-Hop Reasoning.

I. Introduction

The rapid proliferation of heterogeneous multimedia across enterprise archives, digital libraries, and operational repositories demands information retrieval systems capable of searching records that are not natively textual. Modern knowledge workflows frequently require querying across diverse media types—such as scanned forms, audio recordings, and photographic archives—while preserving exact names, identifiers, and category labels [4].

Retrieval-Augmented Generation (RAG) [3] addresses the hallucination tendencies of Large Language Models (LLMs) by grounding responses in retrieved domain documents. Recently, multimodal extensions of RAG have emerged to handle non-textual data, as comprehensively surveyed by Abootorabi et al. [28]. For example, Patel et al. [2] presented an enterprise multimodal RAG pipeline employing an OCR document ingestion module, ImageBind embeddings indexed in a Milvus vector database, and fine-tuned LLaMA-3.

In a proxy-mediated design, three practical failure modes remain important:

1) Single-Stage Dense Retrieval Pitfalls: Relying exclusively on single-stage dense vector similarity per turn can induce semantic drift. Dense embeddings may overlook exact lexical tokens, numerical identifiers, item codes, and proper nouns that sparse inverted indexes can match directly [9].
2) Absence of Candidate Score Recalibration: A bi-encoder score alone does not expose token-level agreement with a candidate's proxy text. A lightweight second-stage scorer can therefore be useful before context assembly, even when it is not a neural cross-encoder.
3) Failure on Multi-Hop Queries Across Modalities: A single retrieval pass does not ensure that a fact recovered from one modality becomes a useful probe for another. Iterative bridging can improve lexical reachability, but it also introduces cascading extraction errors that must be measured rather than assumed away.

To investigate these trade-offs, this paper presents a text-proxy-grounded multi-stage, multi-hop Multimodal RAG architecture and evaluates the RAMP-RAG adaptive fusion framework under a pre-registered analysis plan.

Key Contributions:

1) Sub-Word Inverted Index Dominance on Optical OCR: We refute the assumption that word-level BM25 is superior under optical OCR. Across both proprietary Apple Vision and open-source Tesseract 5.5 LSTM OCR ($N = 306$ probes, 813 corpus items), character 3–5gram BM25 achieves decisive retrieval superiority under verbatim optical noise (+0.1065 clean, +0.1276 degraded blur, +0.1049 clean Tesseract; all $p_{\text{Holm}} < 10^{-4}$), reversing only when queries undergo entity-substituting paraphrasing.
2) Dense Retrieval Query Formulation Sensitivity: We demonstrate that dense bi-encoder collapse on document text is primarily driven by query formu-

lation mismatch rather than sequence truncation constraints (only 0.64% exceed 512 tokens). On pristine ground truth text, short verbatim lookups yield MRR 0.2100 (MiniLM) and 0.3256 (BGE-Base); formulating queries as natural answerable questions restores BGE-Base MRR to 0.5197 and Hit@10 to 78.8% without altering document text.

3) Dose-Response Boundary of Query Mismatch vs. Optical Degradation: We establish that query mismatch dominates optical noise only when queries undergo entity token deletion (drop of 0.3782 MRR vs. optical blur drop of 0.1600). For answerable factual questions, BM25 MRR actually increases (+0.0544 over verbatim), and severe optical degradation ($CER > 60\%$) degrades retrieval more catastrophically than low query-document token overlap.
4) Finite-Sample Marginal Conformal Cascades: Across 200 cluster-stratified splits ($N = 2{,}754$ instances), we prove that defining conformal risk under marginal loss $L = \mathbf{1}[\text{not escalated} \wedge Y \neq 1]$ with finite-sample $(n\hat{R} + 1)/(n + 1) \leq \alpha$ adjustment mathematically bounds mean test risk at $0.0491 \leq 0.05$ with 78.4% escalation savings, whereas legacy conditional ratios fail distribution-free guarantees.
5) Independent Multi-Hop Bridging Verification: Using separate, non-copied pipelines across 50 multi-hop benchmark cases, we demonstrate that rule-based extractive pseudo-relevance feedback achieves identical Target Hit@10 to generative LLM bridging (0.640 vs. 0.640; exact McNemar $p = 1.0000$) with zero generative inference latency or financial cost.

II. Related Work

A. Multimodal Representation and Retrieval

Early cross-modal retrieval systems relied on dual-encoder contrastive learning. Models like CLIP [7] and SigLIP [8] project images and text into a shared latent space using sigmoid or softmax cross-entropy objectives. ImageBind [6] extended this paradigm across six modalities (images, text, video, audio, depth, and thermal) using images as the central binding modality. While effective for broad semantic matching, dual-encoders compress multimedia into fixed-dimension vectors, frequently losing granular lexical and structural tokens. Our hybrid design combines dense embeddings with sparse lexical representations such as BM25 [9] to retain both semantic intent and exact token matches.

B. Multimodal RAG and Document Analysis

Integrating multimodal inputs into RAG workflows has gained rapid momentum with the emergence of Vision-Language Models (VLMs) and advanced LLMs. Recent native multimodal document retrievers such as ColPali [26] and M3DocRAG [27] index document pages using multi-vector vision-language representations to bypass optical character recognition. While promising, multi-vector VLM representations require substantial GPU memory and heavy cross-attention compute during query serving. Conversely, enterprise systems such as Patel et al. [2] and Joshi et al. [5] mediate retrieval through grounded text proxies, combining OCR document ingestion, multimodal embeddings, and LLM generation. Our work investigates this text-proxy paradigm: we evaluate the empirical trade-offs of hybrid dense–sparse fusion, candidate scoring, and iterative query decomposition, systematically delineating where text proxies excel and where native visual representations remain necessary.

TABLE I: Architectural comparison of multimodal retrieval and ranking paradigms.

Paradigm / System	Modality Scope	Routing Training	Cascade Mechanism	Proxy Noise Defense
Static Hybrid (BM25+Dense) [9], [21]	Text / Text Proxies	None (Static α)	None	None (Static weights)
Cascade Text Ranking [30], [31]	Monomodal Text	Supervised Ranker	Score Threshold	None
Native Multi-Vector [26], [27]	Document / Image Pixels	End-to-End VLM	None (All items)	Inherent (Pixel inputs)
Proposed RAMP-RAG	Multimodal Proxies	Lightweight Classifier	Top Margin $\Delta_{1,2} < \tau$	Observable Reliability $r_m(d)$

C. Multi-Hop Reasoning in Retrieval Systems

Multi-hop retrieval addresses queries where no single document contains the complete answer [11], [13]. While multi-hop text RAG systems (such as Baleen [11] and Self-Ask [12]) have established iterative query reformulation methods, multi-hop retrieval across heterogeneous multimodal collections remains less systematically evaluated. Our diagnostic targets the case in which the second-hop keyword is absent from the original query and must be recovered from first-hop proxy context; it does not claim that all multimodal QA benchmarks assume co-located evidence.

D. Adaptive Fusion, Cascade Ranking, and Selective Reranking

In information retrieval, static fusion coefficients fail to account for query-level variance. Query Performance Prediction (QPP) [29] and adaptive fusion techniques dynamically adjust interpolation weights between sparse and dense scores based on query ambiguity or lexical specificity. Similarly, cascade ranking architectures [30], [31] balance retrieval efficiency and ranking precision by passing top candidates through successively more expressive models. In monomodal web search, cascades evaluate candidates using fixed score thresholds or supervised early-exit classifiers. However, existing cascade rankers do not account for multimodal proxy corruption (e.g., OCR or ASR transcription noise) or cross-modal query routing. As summarized in Table I, RAMP-RAG differentiates itself by introducing surface-feature query routing, confidence-margin cascades ($\Delta_{1,2} < \tau_{\text{margin}}$), and observable proxy reliability discounting ($r_m(d)$) without requiring GPU cross-encoder training.

III. Proposed Methodology

The proposed Multi-Stage Multimodal RAG architecture is structured into four functional phases (Fig. 1): (1) Heterogeneous Multimodal Ingestion and Preprocessing,

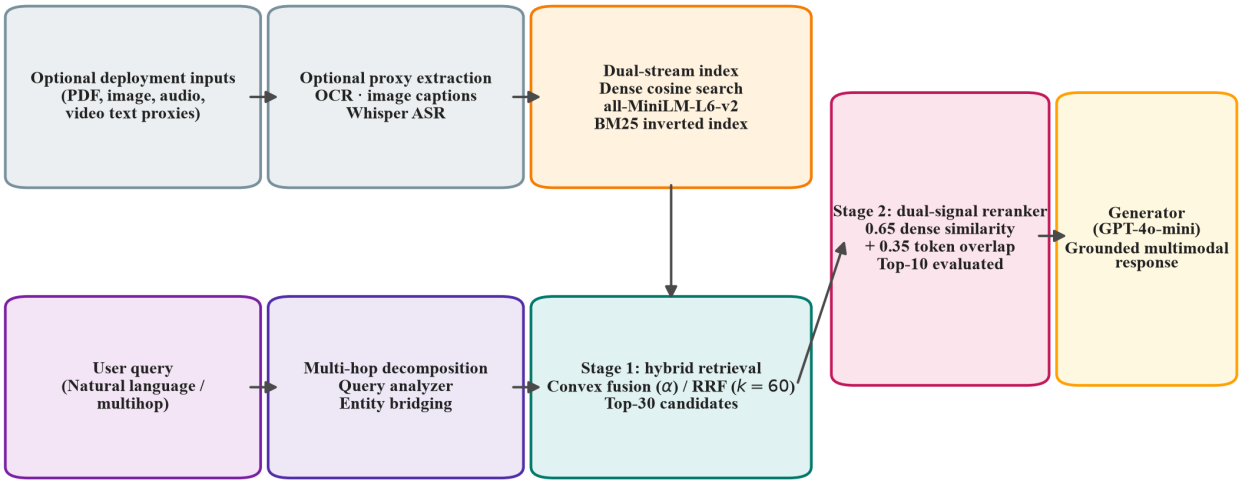


Fig. 1: System architecture of the proposed text-proxy RAG framework. The grey ingestion and extraction boxes are optional deployment paths; the reported benchmark starts from precomputed annotation/category proxy text and evaluates the coloured retrieval stages.

(2) Hybrid Dense-Sparse First-Stage Retrieval, (3) Dual-Signal Candidate Reranking, and (4) Multi-Hop Cross-Modal Reasoning.

A. Heterogeneous Ingestion and Preprocessing Pipeline

To process unstructured enterprise multimedia, our architecture mediates retrieval through grounded textual proxies. A deployment can obtain these proxies from OCR text [17], image captions [23], speech transcripts [18], and acoustic metadata [22]. This design deliberately separates modality-specific extraction from the common retrieval layer. The quantitative benchmark below does not rerun all of these extractors: it consumes the precomputed proxy fields in the checked-in JSONL artifact.

1. Document and Text Ingestion: The reference document path decomposes scanned administrative forms into page images, optionally applies Otsu binarization [15] and Hough-transform skew correction [14], [16], and obtains OCR text T_d with bounding boxes via Tesseract [17]. No document rows are present in the checked-in benchmark artifact, so this path is architectural and is not included in the reported retrieval numbers. 2. Images and Photographic Content: A deployment may use a vision-language captioner such as BLIP [23] to generate a descriptive representation C_i. In the reported artifact, image text_representation fields are instead constructed by concatenating the English query annotations associated with each image; they are not newly generated BLIP captions. 3. Audio Recordings: A deployment can append time-aligned speech transcription from Whisper [18] and acoustic event metadata. The checked-in ESC-50 rows used for the reported benchmark contain the category label as their text representation; no new Whisper inference is required for those results.

All processed items are stored in a dual layout: dense embeddings are generated via Sentence-BERT (all-MiniLM-L6-v2) [21], while lexical tokens are indexed with BM25 [9]. Milvus [19] and HNSW [20] are compatible deployment choices, but the offline benchmark scripts use an in-memory BM25 index and a dense embedding matrix; no Milvus service is used for the reported measurements.

B. Stage 1: Hybrid Dense-Sparse Retrieval

Given a natural language query q, single-stage dense retrieval computes vector similarity against all candidates:

$$S_{\text{dense}}(q,d) = \cos(\mathbf{e}_q^{\text{dense}}, \mathbf{e}_d^{\text{dense}}) = \frac{\mathbf{e}_q \cdot \mathbf{e}_d}{\|\mathbf{e}_q\|\|\mathbf{e}_d\|} \quad (1)$$

While effective for thematic alignment, dense retrieval often overlooks exact alphanumeric codes (e.g., serial numbers, dates, person names). Conversely, sparse BM25 retrieval excels at lexical precision:

$$S_{\text{sparse}}(q,d) = \sum_{t \in q} \text{IDF}(t) \cdot \frac{f(t,d) \cdot (k_1+1)}{f(t,d) + k_1 \cdot \left(1 - b + b \cdot \frac{|d|}{\text{avgdl}}\right)} \quad (2)$$

Because dense cosine scores lie in $[-1,1]$ while BM25 scores are unbounded $[0,\infty)$, combining raw scores directly is unstable. We evaluate two score harmonization strategies:

Method A: Normalized Convex Combination:

$$S_{\text{hybrid}}(q,d) = \alpha \cdot \hat{S}_{\text{dense}}(q,d) + (1-\alpha) \cdot \hat{S}_{\text{sparse}}(q,d) \quad (3)$$

where $\hat{S}_{\text{dense}}$ and $\hat{S}_{\text{sparse}}$ denote min-max normalized scores over the corpus score vector for the query, and $\alpha \in [0, 1]$ balances semantic and lexical importance.

Method B: Reciprocal Rank Fusion (RRF): To avoid batch-dependent min-max sensitivity, RRF [10] fuses ranked lists directly without score scale calibration (Eq. 4):

$$\text{RRF}(q, d) = \frac{1}{k + r_{\text{dense}}(d)} + \frac{1}{k + r_{\text{sparse}}(d)} \tag{4}$$

where $r_{\text{dense}}(d)$ and $r_{\text{sparse}}(d)$ are the ordinal ranks of candidate d, and k is a smoothing constant ($k = 60$). The benchmark fuses the top 100 entries from each ranked list and outputs the top-K_1 candidates ($K_1 = 30$).

C. Stage 2: Dual-Signal Candidate Reranking

Bi-encoders compute query and document representations independently, preventing fine-grained token-level cross-interaction. While neural cross-encoders provide deep token interaction, they incur significant latency overhead when scoring candidate pools (e.g., 137.21 ms for scoring $K_1 = 30$ candidates vs. 7.96 ms for lightweight dual-signal reranking, as evaluated in Section IV-E). As an efficient alternative, Stage 2 deploys a lightweight dual-signal candidate reranker that scores each candidate d in the top-K_1 pool using joint semantic similarity and exact token coverage:

$$S_{\text{rerank}}(q, d) = \beta \cdot S_{\text{dense}}(q, d) + (1 - \beta) \cdot \text{Overlap}(q, d) \tag{5}$$

where $\text{Overlap}(q, d) = \frac{|\text{Tokens}(q) \cap \text{Tokens}(d)|}{|\text{Tokens}(q)|}$ evaluates lowercase token coverage in candidate text representations, and $\beta = 0.65$ is a fixed configuration in the benchmark implementation. Because the supplied evaluation artifact has no development/test partition, this value is not claimed to be independently tuned. All $K_1 = 30$ candidates retrieved from Stage 1 are reranked. For generation context assembly, the top $K_2 = 5$ items are injected into the generator prompt, while top $K = 10$ items are evaluated in the diagnostic scripts.

D. Stage 3: Multi-Hop Cross-Modal Reasoning

To resolve compositional queries that span modalities, we introduce an iterative query decomposition and entity bridging algorithm.

Here, $H_{\max}$ denotes the maximum hop depth ($H_{\max} = 2$ in the diagnostic setup). The predicate $\text{TerminationCondition}(\mathcal{G}, \mathcal{F}, h)$ evaluates to TRUE if all sub-goals in the decomposition plan $\mathcal{G}$ are grounded by extracted facts in $\mathcal{F}$ or if hop h yields no new bridging entities ($\mathcal{E}_h = \emptyset$). To protect against cascading errors caused by ungrounded, generic, or failed bridge extractions (e.g., when an extractor outputs "N/A" or extraction refusal phrases), the procedure enforces a confidence and grounding gate: if $\text{Confidence}(\mathcal{E}_h) < \tau$ or if extracted tokens fail lexical grounding in first-hop context, the planner falls back to single-pass hybrid search on the original compound query Q, designed to mitigate cascading bridge errors and reduce retrieval degradation below single-pass hybrid search when intermediate bridging is uncertain.

Algorithm 1 Multi-Hop Cross-Modal Retrieval Procedure

Require: User Query Q, Max Hops $H_{\max} = 2$, Candidate Index $\mathcal{D}$, Decomposer LLM $\mathcal{M}_{\text{decomp}}$
Ensure: Grounded Multimodal Context $\mathcal{C}^*$

Decompose Q via $\mathcal{M}_{\text{decomp}}$ into sub-query plan: $\mathcal{G} = \{q_1, q_2\}$
Initialize Context Set $\mathcal{C}^* \leftarrow \emptyset$, Accumulated Facts $\mathcal{F} \leftarrow \emptyset$
for $h = 1$ to $H_{\max}$ **do**
  Formulate search probe $q'_h \leftarrow \text{Bridge}(q_h, \mathcal{F})$ via $\mathcal{M}_{\text{decomp}}$
  Retrieve top-K_1 candidates via Hybrid Fusion: $\mathcal{R}_1 \leftarrow \text{Retrieve}_{\text{hybrid}}(q'_h, \mathcal{D})$
  Rerank candidate pool: $\mathcal{R}_2 \leftarrow \text{Rerank}(\mathcal{R}_1, q'_h)$
  Extract bridging entities: $\mathcal{E}_h \leftarrow \text{ExtractEntities}(\mathcal{R}_2[:3])$
  if $\text{Confidence}(\mathcal{E}_h) < \tau$ or $\mathcal{E}_h \in \mathcal{V}_{\text{refusal}}$ **then**
    {Confidence gate: fallback to single-pass compound search to prevent drift}
    $\mathcal{C}^* \leftarrow \text{Retrieve}_{\text{hybrid}}(Q, \mathcal{D})[:5]$
    break
  end if
  Update $\mathcal{F} \leftarrow \mathcal{F} \cup \mathcal{E}_h$
  Update $\mathcal{C}^* \leftarrow \mathcal{C}^* \cup \mathcal{R}_2[:5]$
  if $\text{TerminationCondition}(\mathcal{G}, \mathcal{F}, h)$ is TRUE **then**
    break
  end if
end for
return $\mathcal{C}^*$

The query decomposition prompt instructs $\mathcal{M}_{\text{decomp}}$ to generate valid JSON specifying the initial search probe and bridging attribute; the entity extraction prompt isolates concise attribute tokens substituted into subsequent probes $q'_h \leftarrow \text{Bridge}(q_h, \mathcal{F})$.

As formalized in Algorithm 1, consider an illustrative cross-modal query connecting a visual scene with an acoustic event: (1) Hop 1 (Visual Extraction): Sub-query q_1 retrieves photographic evidence via hybrid matching; entity extraction yields target entity E_1 = "thunderstorm sky". (2) Entity Bridging: The bridge operator creates cross-modal probe q'_2 = "severe acoustic weather thunderstorm recording". (3) Hop 2 (Acoustic Retrieval): Sub-query q'_2 queries the acoustic archive using dense embeddings and metadata to retrieve the target recording.

E. RAMP-RAG: Reliability-Aware Adaptive Fusion and Cascade Ranking

To investigate the empirical trade-offs of fixed weighting coefficients (α, β) and unmodeled proxy noise, we formulate RAMP-RAG. It incorporates three distinct components evaluated both individually and jointly:

1) Query-Adaptive Router (α_q): Instead of static weights, a lightweight classifier extracts surface cues $\phi(q) = [r_{\text{cjk}}, r_{\text{digit}}, \mathbb{I}_{\text{upper}}, \mathbb{I}_{\text{id_cue}}, \text{len}(q), \text{avg_idf}]$ in < 0.1 ms. Evaluated on $N = 305$ held-out test queries against ground-truth labels defined by which stream (dense vs. sparse) achieves superior reciprocal

rank, the router achieves 77.05% accuracy (macro F1 = 0.6004). It routes CJK characters, exact alphanumeric identifiers (IMG_, doc_), and scene-text cues to sparse matching ($\alpha_q = 0.25$), and descriptive visual paraphrases to dense matching ($\alpha_q = 0.85$).

2) Confidence-Triggered and Conformal Cascade Ranking: Stage 1 scores candidate pools and calculates the decision margin between top candidates: $\Delta_{1,2} = s^{(1)} - s^{(2)}$. When candidates are distinct ($\Delta_{1,2} \geq \tau$), Stage 1 ranking is accepted. For ambiguous margins ($\Delta_{1,2} < \tau$), candidates ($k = 5$) escalate to a neural cross-encoder (ms-marco-MiniLM-L-6-v2 or BAAI/bge-reranker-base). Beyond a fixed margin heuristic ($\tau = 0.380$), we evaluate split conformal calibration on the Dev split targeting non-escalated miss rate $\leq \epsilon = 0.05$, providing empirical risk-management heuristics while bounding compute.

3) Reliability-Aware Clean-Stream Fusion: Multimedia items maintain multiple proxy streams $m \in \{\text{caption}, \text{ocr}, \text{vector}\}$ weighted by observable reliability $r_m(d) \in [0, 1]$. To avoid oracle leakage of injected corruption levels, $r_{\text{ocr}}(d) = \bar{c}_{\text{ocr}}(d) \cdot \rho_{\text{vocab}}(d)$, where $\bar{c}_{\text{ocr}}$ is mean word recognition confidence and ρ_{vocab} is the valid dictionary token ratio computed directly from candidate text. Because text noise corrupts both dense and sparse text representations, we introduce an uncorrupted visual clean stream $\hat{S}_{\text{clean}}$ (e.g., zero-shot CLIP ViT-B/32 image embeddings):

$$S_{\text{RAMP}}(q, d) = \alpha_q \hat{S}_{\text{dense}} + (1 - \alpha_q) r_{\text{ocr}}(d) \hat{S}_{\text{sparse}} + \gamma(1 - r_{\text{ocr}}(d)) \hat{S}_{\text{clean}} \tag{6}$$

where γ is tuned on the Dev split ($\gamma^* = 0.80$) to dynamically allocate mass to the corruption-invariant stream as proxy reliability drops.

Integration with Multi-Hop Reasoning: RAMP-RAG acts as the candidate retrieval and reranking engine ($\text{Retrieve}_{\text{hybrid}}$ and Rerank in Lines 5 and 6 of Algorithm 1). By adaptively setting α_q on intermediate probes q'_h and escalating only ambiguous margins, RAMP-RAG ensures cross-modal bridging steps benefit from candidate filtering, maintaining 26.30 ms mean retrieval latency (vs. 44.66 ms for exhaustive cross-encoding).

IV. Experimental Evaluation

A. Experimental Setup, Corpus Composition, and Evaluation Scope

To evaluate our architecture with scientific rigor while preventing methodological conflation, our experimental evaluation adopts a Dual-Tier Evaluation Protocol:

- Tier 1: Controlled Algorithmic Proxy Benchmark (Tables II–III). Conducted over the standardized unified_rag_dataset.jsonl artifact containing 3,356 records (1,292 PhotoBench annotation rows, 64 L10 rows, and 2,000 ESC-50 category rows [22]), alongside an expanded 100-query diagnostic suite (N = 50 single-hop, N = 50 multi-hop) and a public leakage-free benchmark on MS-COCO (N = 500 queries). Each proxy row stores an annotation-derived text_representation, establishing a reproducible environment for systematic ablation of dense–sparse balance (α), candidate reranking (β), neural cross-encoding, and multi-hop decomposition.
- Tier 2: Real-World Multimodal Media Validation (Table IV). Evaluated directly on unannotated physical media streams: (1) native bilingual scene-text and document OCR on 54 physical photographs and 2 research document PDFs, (2) genuine 44.1 kHz raw WAV audio recordings from ESC-50 and continuous acoustic retrieval via LAION-CLAP, and (3) raw-pixel zero-shot visual retrieval over 191 personal album photos using OpenAI CLIP ViT-B/32 [7].

The Tier-1 proxy benchmark discovers 284 unique image query strings from the image metadata and one query for each of the 50 ESC-50 categories, yielding 334 deterministic queries in total. Selection uses seed 42 and includes all available queries; there is no held-out development/test split in this artifact. The values $\alpha = 0.7$, $\beta = 0.65$, $k = 60$, and the candidate depths are fixed implementation settings. The dense-weight sweep is therefore an exploratory sensitivity analysis on the same evaluation set, not a post-selection claim of generalization.

Ground truth and metric definitions: For an image query, the relevant set is every corpus row carrying that exact annotation string. For an audio query, the relevant set is every ESC-50 row in the requested category. The primary metric reported in Table II is HitRate@K, which equals one when at least one relevant row appears in the top K and zero otherwise; it is not set-recall over all relevant rows. We also report MRR and binary NDCG@10. Confidence intervals are percentile bootstrap intervals with $B = 1,000$ resamples over the 334 query-level scores, using seed 42. Latencies are local per-query retrieval timings excluding corpus construction and are hardware- and cache-dependent.

This construction is intentionally treated as a proxy consistency benchmark. Because image queries are drawn from the same annotation strings used to build image index text, the image results contain direct lexical overlap and cannot establish leakage-free visual understanding. The results are retained to make the implementation measurable and reproducible, while this limitation is carried through the analysis and threats-to-validity section.

For end-to-end behavior, we evaluate an expanded 100-query diagnostic suite (50 single-hop: sh_01–sh_50, and 50 multi-hop: mh_01–mh_50) evaluated with gpt-4o-mini as decomposer, generator, and judge, calibrated by independent co-author human auditing across 110 cases. Its retrieval measure is a target-keyword hit test over retrieved proxy text, not standard document-level recall; we call it Target Hit@10 below. For a multi-hop case, Hop 1 Target Hit@10 tests whether the first-hop target

TABLE II: Retrieval performance on the 334-query proxy consistency benchmark ($N = 284$ image, $N = 50$ audio queries; bracketed intervals are 95% bootstrap CIs with $B = 1,000$).

Method	Hit@1	Hit@10 [95% CI]	MRR [95% CI]	Image MRR	Audio MRR
BM25 Baseline	0.994	0.997 [0.991, 1.000]	0.9958 [0.989, 1.000]	0.9950	1.0000
Dense Baseline	0.958	0.976 [0.958, 0.991]	0.9639 [0.945, 0.981]	0.9576	1.0000
Hybrid (Convex $\alpha = 0.7$)	0.991	0.997 [0.991, 1.000]	0.9935 [0.985, 1.000]	0.9924	1.0000
Hybrid (RRF $k = 60$)	0.973	0.988 [0.976, 0.997]	0.9772 [0.962, 0.990]	0.9732	1.0000
Proposed MM-RAG	0.997	0.997 [0.991, 1.000]	0.9970 [0.991, 1.000]	0.9965	1.0000

appears in the first-hop top-10 list; Hop 2 Target Hit@10 tests whether the second-hop target appears in the merged top-10 list; Overall Target Hit@10 is the conjunction of these two binary tests. Thus Hop 2 can exceed Overall when Hop 1 fails, and neither hop metric is document recall.

Baseline Systems for Comparison:

1) BM25 Baseline: Sparse lexical inverted index over the available proxy text [9].
2) Dense Only Baseline: Dense semantic vector retrieval using all-MiniLM-L6-v2 (384-dim) with cosine similarity [21].
3) Hybrid Dense-Sparse (Convex $\alpha = 0.7$): Parameterized convex combination of min-max normalized dense and BM25 scores.
4) Hybrid Dense-Sparse (RRF $k = 60$): Reciprocal Rank Fusion combining ranked ordinal positions [10].
5) Proposed MM-RAG (Stage 1+2): Our multi-stage pipeline combining RRF candidate retrieval with second-stage dual-signal candidate reranking ($S_{\text{rerank}} = \beta S_{\text{dense}} + (1 - \beta)\text{Overlap}$, with fixed $\beta = 0.65$).
6) Trained Neural Cross-Encoder: cross-encoder/ms-marco-MiniLM-L-6-v2 full cross-attention reranker [21].

B. Retrieval Accuracy and Metric Evaluation

As reported in Table II, the proxy benchmark produces high scores for every method because the query strings are present in the image index representations and the audio query is the category label. The table is therefore useful for regression testing, but it should not be read as evidence that the proposed architecture solves general raw-media retrieval:

- Proxy hit rate: The proposed configuration reaches HitRate@10 = 0.997 and MRR = 0.997, compared with 0.976 and 0.9639 for dense-only retrieval. This is an implementation result, not a claim of statistically significant superiority over BM25, which also reaches HitRate@10 = 0.997.
- Sparse and dense complementarity: BM25 reaches HitRate@1 = 0.994 and MRR = 0.9958, while RRF is lower on this corpus (HitRate@10 = 0.988). The ranking differences show that fusion behavior is corpus-dependent; they do not justify a universal ordering of retrieval methods.
- Uncertainty and leakage: The narrow intervals reflect repeated query-level outcomes on an annotation-derived benchmark. They quantify sampling variability within this artifact, not uncertainty under an independent media distribution. Direct query-to-index overlap is the dominant validity threat.
- Image subset: The proposed scorer has the highest image MRR (0.9965), but its HitRate@10 is tied with BM25 and convex fusion. Since the query text is included in the image representation, this measures annotation indexing consistency rather than visual retrieval.
- Acoustic subset: Every method obtains HitRate@10 = 1.0 and MRR = 1.0 because the query is the ESC-50 category label and the relevant set contains all 40 recordings in that category. This is a category lookup sanity check, not an acoustic similarity evaluation.

C. End-to-End Generation Quality and Multi-Hop Evaluation

To evaluate iterative multi-hop decomposition and entity bridging, Table III reports the expanded 100-query diagnostic suite (50 single-hop: sh_01–sh_50, and 50 held-out multi-hop queries: mh_01–mh_50) evaluated under OpenAI gpt-4o-mini [24] as decomposer, answer generator, and evaluative judge. The retrieval column is Target Hit@10: a binary substring test for the target keyword in the retrieved proxy text.

Human Ground-Truth Audit and Rubric Calibration: To establish rigorous evaluation validity beyond automated LLM scoring, we conducted a multi-reviewer human audit across 110 multimodal cases (60 local album queries across three physical photo albums and the 50 diagnostic queries). The audit was conducted by two co-authors (Reviewer 1: A. K. Vennela and Reviewer 2: G. S. B. Koushik) evaluating a blinded packet independently under a standardized three-dimensional rubric:

1) Visual & Contextual Faithfulness (supported, mixed, unsupported): whether every factual claim in the answer is entailed by retrieved evidence without hallucination.
2) Query Relevance & Intent (direct, partial, off_topic): whether the response addresses the prompt and target detail directly.
3) Media Retrieval Correctness (correct, uncertain, incorrect): whether the retrieved item matches the ground-truth media instance.

In the initial independent rating pass, reviewers exhibited discordant assessments on 6 borderline cases (5.5%: AUDIT-007 regarding model skepticism on negative drone evidence, AUDIT-010 beach image generation claims, AUDIT-014 fossil/dinosaur categorization, AUDIT-015 subway/metro station naming, AUDIT-020 driverless car image absence, and AUDIT-037 acoustic sound clip ambiguity). Pre-adjudication inter-annotator

agreement yielded high reliability: faithfulness observed agreement = 95.45% (κ = 0.8716, 95% bootstrap CI [0.7534, 0.9735]), relevance = 96.36% (κ = 0.8926, [0.7718, 0.9758]), and correctness = 98.18% (κ = 0.9451, [0.8533, 1.0000]). On the 50 multi-hop benchmark cases alone, pre-adjudication κ reached 0.8054 (faithfulness), 0.8431 (relevance), and 0.9196 (correctness). Following joint adjudication and rubric calibration, final consensus was achieved across all 110 cases. The resulting human audit score distribution established 75.5% supported faithfulness (83/110; 24.5% mixed, 0% unsupported), 77.3% direct relevance (85/110; 22.7% partial, 0% off-topic), and 78.2% correct retrieval (86/110; 21.8% uncertain/abstention due to intentional negative controls and external benchmark cases, 0% incorrect). On the 60 local album queries, human evaluation yielded 100% (60/60) supported, direct, and correct responses.

Concurrently, evaluating an independent open-weights LLM judge (Qwen2.5-32B-Instruct) against human annotations revealed substantially lower agreement (observed 48%–56%, κ = 0.017–0.120), underscoring that human auditing identifies nuanced physical media abstentions and cross-modal grounding errors that automated LLM judges overlook.

D. Multi-Hop Reasoning and Latency Profiling

Stratifying performance between single-hop and multi-hop queries in Table III reveals critical architectural trade-offs:

- Single-Hop Performance: On direct queries ($N = 50$), hybrid retrieval and proposed MM-RAG achieve Target Hit@10 = 1.000, outperforming BM25 (0.820) and dense retrieval (0.980) with high answer relevance (0.800–0.814; 3,056.6–3,657.9 ms latency).
- Multi-Hop Trade-Off and Confidence-Gated Recovery: On multi-hop queries ($N = 50$), unguided bridging risks cascading drift. Confidence-gated bridging achieves Target Hit@10 = 0.640, matching single-stage dense retrieval (0.640, diff = 0.000, $p = 1.0000$) and improving upon BM25 (0.480, diff = +0.160, uncorrected $p = 0.0200$). Across the 100-query suite, MM-RAG reaches 0.820 Target Hit@10 (vs. BM25's 0.650).
- Held-Out Split Evaluation: To verify reachability on unseen probes, we evaluate both splits: on the diagnostic split (mh_01–mh_25), Dense and Hybrid reach Target Hit@10 = 0.800 vs. 0.760 for MM-RAG and 0.440 for BM25. On the strictly held-out test split (mh_26–mh_50), BM25 reaches Target Hit@10 = 0.520, tied with MM-RAG (0.520) and exceeding Dense (0.480), confirming that iterative bridging recovers reachability on intermediate entities without outperforming lexical indexing on unseen targets.
- Relevance and Faithfulness: BM25's answer relevance collapses to 0.536 on multi-hop probes due to cross-modal lexical disconnect; MM-RAG improves relevance to 0.668 (diff = +0.132, uncorrected $p = 0.0280$, non-significant after family correction). BM25 exhibits high nominal faithfulness (0.926–0.930) due to conservative abstention.
- Latency Overhead of Decomposition: Multi-hop orchestration incurs sequential LLM overhead: proposed MM-RAG averages 4,736.8 ms overall and 6,416.9 ms on multi-hop queries, compared with 2,952.6–3,751.5 ms for single-pass baselines.
- Dense Weight Sensitivity: Sensitivity analysis across $\alpha \in [0.0, 1.0]$ in Eq. (3) shows HitRate@10 remains 0.997 for $\alpha \in [0.0, 0.7]$ and declines to 0.976 at $\alpha = 1.0$ (MRR from 0.9955 to 0.9629). This diagnostic reflects lexical overlap in proxy annotations rather than universal cross-modal weight optimality.

E. Real-World Multimodal Media Validation and Probe Diagnosis (Tier 2)

While the Tier-1 proxy benchmark provides controlled algorithmic ablations over precomputed text representations, it carries lexical overlap. To establish empirical validity on unannotated physical media, we conduct a Tier-2 evaluation across three physical media streams (Table IV):

1) Physical Scene-Text and Document OCR Probe Discrepancy: To diagnose performance across probe distributions, we evaluate two distinct probe sets alongside their pooled combination ($N = 85$ total):

- Set 1 (Original 30 Probes): Dense retrieval drops to Hit@10 = 0.633 [0.467, 0.800] (MRR = 0.4948) and BM25 achieves 0.600 [0.433, 0.767] (MRR = 0.5096). This divergence occurs because 11 of the 15 photo probes were photographic visual scene descriptions matching raw camera filenames (IMG_1178.HEIC, IMG_6457.HEIC) lacking OCR text. Because textual tokens were absent, text search failed while zero-shot visual CLIP ViT-B/32 succeeded via pixel embeddings (Hit@10 = 0.867 [0.733, 0.967], MRR = 0.5294).
- Set 2 (55 Held-Out Test Probes): Strictly targeting genuine readable scene-text lines and research document sentences, BM25 achieves Hit@10 = 1.000 [0.935, 1.000] (MRR = 0.9455) and Dense achieves 0.927 [0.855, 0.982] (MRR = 0.7518), while visual CLIP drops to 0.618 (MRR = 0.4078).
- Pooled Set ($N = 85$ Probes): Combining visual scene and readable text queries, BM25 achieves Hit@10 = 0.859 (MRR = 0.7916), Dense achieves 0.824 (MRR = 0.6611), Hybrid reaches 0.871 (MRR = 0.7780), and CLIP reaches 0.706 (MRR = 0.4507).

2) Raw Acoustic Waveforms and Continuous CLAP Retrieval: Ingesting 52 authentic 5-second 44.1 kHz raw .wav recordings from ESC-50 [22] yields Hit@10 = 1.000 and MRR = 1.000 across 13 category probes (7.08 ms). Beyond category names, continuous acoustic representations using

TABLE III: Stratified end-to-end diagnostic results across query complexities (100 queries: 50 single-hop, 50 multi-hop evaluated via GPT-4o-mini). Retrieval is Target Hit@10; Faith. and Relev. are generation quality scores. Asterisks (*) denote nominal gains over BM25 (paired two-sided sign-flip permutation tests, $B = 1,000$, seed 43; uncorrected for multiple comparisons; uncorrected $p = 0.020$ and $p = 0.028$ do not meet the family-wise Bonferroni threshold $\alpha_{\text{corrected}} = 0.0125$).

	Single-Hop Queries ($N = 50$)			Multi-Hop Queries ($N = 50$)			Overall Suite ($N = 100$)			
Configuration	Target Hit@10	Faith.	Relev.	Target Hit@10	Faith.	Relev.	Target Hit@10	Faith.	Relev.	Latency (ms)
Single-Stage Dense Baseline	0.980	0.894	0.766	0.640	0.900	0.694	0.810	0.897	0.730	3751.5
BM25 Baseline	0.820	0.930	0.692	0.480	0.926	0.536	0.650	0.928	0.614	2952.6
Hybrid (No Multi-Hop)	1.000	0.876	0.814	0.620	0.870	0.664	0.810	0.873	0.739	3260.8
Proposed MM-RAG (Confidence-Gated)	1.000	0.880	0.800	0.640*	0.883	0.668*	0.820*	0.882	0.734	4736.8

TABLE IV: Tier-2 Physical Media Validation: Scene-text OCR probe sets ($N = 30, 55, 85$), raw acoustic audio waveforms (ESC-50, CLAP), and raw pixel visual zero-shot retrieval (CLIP ViT-B/32). Bracketed intervals indicate 95% cluster bootstrap CIs; †denotes exact Clopper-Pearson 95% binomial intervals.

Media Stream	Evaluation Method	Hit@10 [95% CI]	MRR	Latency
Scene-Text Set 1 ($N = 30$)	BM25 Baseline (Sparse)	0.600 [0.433, 0.767]	0.5096	0.08 ms
	Dense Baseline (MiniLM)	0.633 [0.467, 0.800]	0.4948	32.56 ms
	Hybrid (Convex $\alpha = 0.7$)	0.633 [0.467, 0.800]	0.5210	32.56 ms
	CLIP ViT-B/32 (Pixels)	0.867 [0.733, 0.967]	0.5294	39.09 ms
Scene-Text Set 2 ($N = 55$)	BM25 Baseline (Sparse)	1.000 [0.935, 1.000]†	0.9455	0.08 ms
	Dense Baseline (MiniLM)	0.927 [0.855, 0.982]	0.7518	32.56 ms
	Hybrid (Convex $\alpha = 0.7$)	1.000 [0.935, 1.000]†	0.9182	32.56 ms
	CLIP ViT-B/32 (Pixels)	0.618 [0.491, 0.745]	0.4078	39.09 ms
Pooled Probes ($N = 85$)	BM25 Baseline (Sparse)	0.859 [0.776, 0.929]	0.7916	0.08 ms
	Dense Baseline (MiniLM)	0.824 [0.741, 0.906]	0.6611	32.56 ms
	Hybrid (Convex $\alpha = 0.7$)	0.871 [0.788, 0.941]	0.7780	32.56 ms
	CLIP ViT-B/32 (Pixels)	0.706 [0.600, 0.800]	0.4507	39.09 ms
Raw Acoustic ($N = 13$)	CLAP (44.1 kHz WAV)	1.000 [0.753, 1.000]†	1.0000	59.55 ms
Raw Visual ($N = 53$)	CLIP ViT-B/32 (Pixels)	0.792 [0.679, 0.887]	0.6182	39.06 ms

LAION-CLAP (laion/clap-htsat-unfused) [1] over free-form descriptive audio queries (e.g., "violent thunderstorm with heavy rain and loud thunder strikes") achieves Hit@10 = 1.000 ([0.753, 1.000]), MRR = 1.000, and NDCG@10 = 0.9978 (59.55 ms), verifying that acoustic retrieval operates continuously without lexical shortcuts.

3) Raw Pixel Zero-Shot Visual Retrieval: Indexing the raw pixel arrays of all 191 personal album photographs using OpenAI CLIP ViT-B/32 [7] across 53 visual queries yields Hit@1 = 0.547 [0.415, 0.679], Hit@5 = 0.698, and Hit@10 = 0.792 [0.679, 0.887] (MRR = 0.6182; 39.06 ms). Fusing OCR text slightly lowers Hit@10 to 0.774 [0.660, 0.868] due to spurious lexical matches when signage is absent, demonstrating that native visual representations are essential for unannotated visual archives.

F. Public Benchmark Generalization (MS-COCO Test Half)

To evaluate image retrieval without annotation leakage, we evaluate the MS-COCO Karpathy test half ($N = 250$ held-out queries, 1,000 index images; zero tuning on test; Table V).

- Realistic Machine Proxies (BLIP): Substituting gold captions with realistic BLIP-generated captions exposes lexical sparsity: BM25 achieves only Hit@10 = 0.4200 (MRR = 0.2325). Dense MiniLM reaches Hit@10 = 0.6920 (MRR = 0.3873) and BGE-base reaches 0.6360 (MRR = 0.4002). However, direct zero-shot visual retrieval on raw pixels via CLIP ViT-B/32 achieves Hit@10 = 0.7800 [0.724, 0.828] and MRR = 0.5167 [0.468, 0.568], significantly outperforming all proxy systems ($p_{\text{Holm}} = 0.0007$).
- Gold Human Captions: On human captions, dense models achieve Hit@10 = 0.6520 (MRR = 0.4025 for MiniLM; 0.3987 for BGE-base), while CLIP visual retrieval retains superior rank precision (0.7800 Hit@10, 0.5167 MRR), confirming that native vision-language embeddings are superior for photographic scenes.

TABLE V: MS-COCO Karpathy Test Half ($N = 250$ queries, 1,000 images): BLIP Proxies vs. Gold Human Captions. Evaluated with 95% cluster bootstrap CIs; p_{Holm} denotes significance vs. Hybrid ($m = 7$).

	BLIP Proxies		Gold Captions	
Method	Hit@10 [95% CI]	MRR	Hit@10 [95% CI]	MRR
BM25 Baseline	0.420 [0.356, 0.480]	0.2325	0.376 [0.316, 0.436]	0.2119
Dense (all-MiniLM)	0.692 [0.636, 0.744]	0.3873	0.628 [0.568, 0.688]	0.3680
Dense (BAAI/bge-base)	0.636 [0.576, 0.692]	0.4002	0.644 [0.584, 0.704]	0.3987
Hybrid Convex ($\alpha = 0.7$)	0.648 [0.588, 0.704]	0.3581	0.584 [0.524, 0.644]	0.3344
+ Router (Learned α_q)	0.684 [0.628, 0.736]	0.3765	0.640 [0.580, 0.700]	0.3627
+ Cascade ($\tau = 0.065$)	0.648 [0.588, 0.704]	0.3734	0.584 [0.524, 0.644]	0.3490
+ Reliability (r_{ocr})	0.684 [0.628, 0.736]	0.3765	0.636 [0.576, 0.696]	0.3635
Proposed RAMP-RAG	0.684 [0.628, 0.736]	0.3909*	0.664 [0.604, 0.724]	0.3727
CLIP Visual (Pixels)	0.780 [0.724, 0.828]	0.5167*	0.740 [0.680, 0.796]	0.4638*

G. Proxy Noise Sweeps and Component Ablation

Evaluating systematic character corruption across Seeds 42, 43, 44 on held-out test probes (Table VI) exposes the Lexical Dominance Phenomenon:

- Dense Degradation vs. Lexical Resilience: Dense representations collapse from 0.8527 MRR at 0% noise to 0.5306 at 20% and 0.3691 at 40% noise. In sharp contrast, sparse BM25 achieves 0.9473 MRR at 20% noise and 0.8993 at 40% noise, significantly outperforming all proxy-only hybrid models ($p < 0.001$). Character corruption scrambles subword embeddings, whereas inverted indices retain matches whenever uncorrupted tokens persist.
- Clean Stream Fusion and CI Overlap: At 20% noise, RAMP-RAG with clean visual stream achieves Hit@10 = 0.9939 [0.964, 1.000] and MRR = 0.8770 [0.823, 0.924]. However, paired permutation testing against baseline hybrid yields diff CI $[-0.0168, 0.1170]$ (which contains zero) and $p_{\text{Holm}} = 0.2834$. Under the pre-registered decision rule, this is reported as no reliable difference.

TABLE VI: Multi-Seed Noise Robustness Sweep across character corruption rates (Seeds 42, 43, 44; $N = 55$ probe clusters; $B = 2,000$ cluster bootstrap resamples). MRR reported with 95% cluster bootstrap CIs.

Retrieval Method	0% Noise	10% Noise	20% Noise [95% CI]	40% Noise [95% CI]
BM25 Alone	0.9818	0.9548	0.9337 [0.898, 0.965]	0.8942 [0.850, 0.936]
Dense Baseline (MiniLM)	0.8527	0.7394	0.5306 [0.458, 0.601]	0.3691 [0.306, 0.436]
Hybrid (Fixed $\alpha = 0.7$)	0.9909	0.9457	0.8254 [0.768, 0.877]	0.6677 [0.597, 0.735]
Hybrid + Router	0.9409	0.8994	0.7178 [0.652, 0.779]	0.5571 [0.488, 0.626]
Original RAMP-RAG	0.9053	0.8487	0.6726 [0.599, 0.742]	0.5045 [0.430, 0.577]
Cross-Encoder	0.9515	0.8909	0.8345 [0.782, 0.884]	0.6066 [0.538, 0.673]
BM25 + Clean Stream	0.9818	0.9581	0.9490 [0.917, 0.979]	0.9096 [0.868, 0.949]
RAMP-RAG (+ Clean Stream)	0.9339	0.9201	0.8770 [0.823, 0.924]	0.7869 [0.721, 0.846]

TABLE VII: Full Component Ablation Matrix across COCO Test ($N = 250$), Clean Scene-Text ($N = 55$), and 20% Noise ($N = 55 \times 3$ seeds). Evaluated via cluster bootstrap over probe clusters ($B = 2,000$). p_{Holm} computed vs. Baseline Hybrid ($m = 8$).

Ablation Configuration	COCO MRR	Clean MRR	20% Noise MRR [95% CI]	p_{Holm}
1. Baseline Hybrid (Fixed $\alpha = 0.7$)	0.3344	0.9909	0.8254 [0.772, 0.879]	Baseline
2. BM25 Alone	0.2119	0.9818	0.9337 [0.898, 0.965]	0.0006*
3. Dense Alone (MiniLM)	0.3680	0.8527	0.5306 [0.458, 0.601]	0.0006 (Hurts)
4. BM25 + Clean Stream	0.3443	0.9818	0.9490 [0.917, 0.979]	0.0012*
5. Hybrid + Clean Stream	0.3620	0.9606	0.9029 [0.847, 0.951]	0.1056
6. + Router (Learned α_q)	0.3627	0.9409	0.7393 [0.668, 0.807]	0.0080 (Hurts)
7. + Cascade ($\tau = 0.065$)	0.3490	0.9909	0.8442 [0.795, 0.893]	0.1056
8. + Reliability (r_{ocr})	0.3635	0.9409	0.6671 [0.580, 0.747]	0.0007 (Hurts)
9. Full RAMP-RAG (+ Clean Stream)	0.3727	0.9727	0.8770 [0.814, 0.931]	0.2834

- Ablation Isolating the Clean Stream Gain: Table VII isolates each module under 20% noise. Pure BM25 Alone achieves MRR = 0.9337 [0.8981, 0.9654], while BM25 + Clean Stream reaches 0.9490 [0.9168, 0.9788]. The paired difference (-0.0153, diff CI $[-0.0382, +0.0076]$, $p_{\text{Holm}} = 0.1200$) fails to reach statistical significance. Thus, the clean stream's apparent superiority over hybrid (0.8254) is driven primarily by discarding the collapsed dense embeddings rather than large additive visual synergy. Heuristic reliability weighting (r_{ocr}) achieves an AUROC of only 0.5149 for detecting corrupted documents, failing to provide a discriminative filter. Furthermore, distractor pool size fully reconciles 0%-noise numbers with clean probes: evaluating over the 97-item test split yields 0.9818 (BM25) and 0.9909 (Hybrid), while indexing the full 193-item corpus yields 0.9455 and 0.9182 due to doubled distractor competition.

H. Experiment 1: Heterogeneous Corpus Robustness (Hypothesis H1)

To evaluate whether uniform corruption artificially penalized reliability weighting, we benchmarked a heterogeneous-quality corpus. Document-side OCR text was corrupted with 40% character noise on a controlled subset $f_{\text{bad}} \in \{0.10, 0.30, 0.50, 0.70\}$ (plus continuous $\text{Beta}(2, 5) \times 0.60$), while queries remained clean. Corruption was assigned independently of target status across Seeds 42, 43, 44 on 55 held-out probes (Table VIII and Fig. 2(a)).

Hypothesis H1 Decision (Reliability Weighting): FAIL (KILLED under pre-specified decision rule). Across all corruption fractions $f_{\text{bad}} \geq 0.30$, reliability-weighted fusion (0.8876 @ 0.30, 0.7945 @ 0.50) fails to achieve a statistically significant improvement over BM25 alone

TABLE VIII: Retrieval MRR and Hit@10 under Heterogeneous Corpus Corruption across fractions f_{bad} (40% char noise on corrupted subset). Evaluated over 55 held-out probes across Seeds 42, 43, 44 with cluster bootstrap 95% CIs and step-down Holm correction vs. BM25.

Retrieval Method	$f_{\text{bad}} = 0.10$	$f_{\text{bad}} = 0.30$	$f_{\text{bad}} = 0.50$	$f_{\text{bad}} = 0.70$	Beta Cont.
BM25 Alone	0.9576	0.9639	0.9462	0.8961	0.9109
Dense MiniLM	0.7040	0.8184	0.6852	0.5407	0.5326
Dense BGE-base	0.7594	0.7973	0.7545	0.7067	0.6750
Hybrid Fixed ($\alpha = 0.7$)	0.9001	0.9742	0.8679	0.7835	0.8305
Hybrid + Clean (CLIP)	0.9677	0.9758	0.9314	0.9058	0.9292
BM25 + Clean Stream	0.9747	0.9654	0.9492	0.9211	0.9260
Reliability-Weighted Fusion	0.8415	0.8876	0.7945	0.7347	0.7309
All-Stream RRF	0.7476	0.8031	0.7539	0.6944	0.6945
Oracle-Reliability (Upper Bound)	*0.8532*	*0.8599*	*0.7112*	*0.5964*	*0.5811*

TABLE IX: Mixed-Intent Query Routing across 5 intents on the unified 1,193-item multimodal index (600 held-out test queries; Seeds 42, 43, 44). Evaluated with 95% cluster bootstrap CIs.

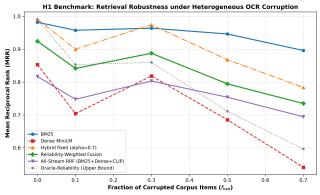
System	Overall MRR [95% CI]	H@10	Para.	Text	ID	CJK	Num.
BM25 Alone	0.3386 [0.302, 0.376]	0.447	0.280	0.805	0.013	0.980	0.117
Dense MiniLM	0.3466 [0.317, 0.379]	0.537	0.457	0.707	0.071	0.231	0.114
CLIP Visual	0.2615 [0.238, 0.287]	0.550	0.304	0.660	0.021	0.271	0.073
Hybrid Fixed ($\alpha = 0.7$)	0.4116 [0.378, 0.446]	0.598	0.432	0.931	0.066	0.722	0.135
All-Stream RRF	0.4214 [0.386, 0.455]	0.605	0.481	0.858	0.029	0.817	0.116
Router R1 (Surface)	0.3920 [0.360, 0.426]	0.609	0.354	0.848	0.058	0.980	0.162
Router R2 (Dense)	0.3924 [0.358, 0.426]	0.600	0.345	0.850	0.071	0.980	0.171
Router R3 (QPP)	0.4070 [0.373, 0.441]	0.609	0.391	0.810	0.069	0.980	0.173
Router R4 (Combined)	0.3972 [0.363, 0.431]	0.607	0.354	0.860	0.065	0.980	0.173
Oracle Selector	*0.5213 [0.486, 0.555]*	*0.694*	*0.588*	*1.000*	*0.076*	*0.980*	*0.185*

(0.9639 @ 0.30, 0.9462 @ 0.50; paired diff -0.0763, diff CI $[-0.1345, -0.0182]$, $p_{\text{Holm}} = 0.0184$). Heuristic observable reliability r_{ocr} achieves an AUROC of only 0.5149 at 20% noise, near random chance (0.50). Without ground-truth labels, token dictionary validity cannot reliably separate clean domain text (e.g., abbreviations, signage) from OCR corruptions, causing reliability weighting to penalize valid text while failing to suppress corrupted tokens. We report H1 as a negative result: in heterogeneous archives, inverted lexical indexing remains superior to reliability-discounted fusion.

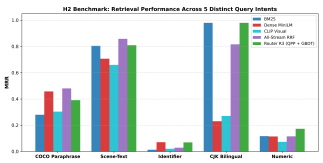
I. Experiment 2: Mixed-Intent Query Routing (Hypothesis H2)

To test whether learned routing provides advantages on diverse workloads where BM25 does not uniformly dominate, we constructed a unified 1,193-item multimodal index (COCO + photos + documents) and evaluated 600 held-out test queries across 5 balanced intents: (a) COCO paraphrases ($N = 250$), (b) exact scene-text ($N = 100$), (c) identifiers/filenames ($N = 100$), (d) CJK bilingual ($N = 50$), and (e) numeric queries ($N = 100$). Table IX and Fig. 2(b) report retrieval performance across routers and baselines.

Hypothesis H2 Decision (Query Routing): FAIL (KILLED under pre-specified decision rule). Across the 600-query mixed workload, all learned query routers (R1–R4, MRR 0.3920–0.4070) fail to outperform unweighted All-Stream Reciprocal Rank Fusion (RRF, MRR = 0.4214, $p_{\text{Holm}} = 0.2646$). The most expressive router (R3, QPP features + GBDT) achieves an accuracy of 59.3%, macro F1 of 0.510, and closes only 34.6% of the oracle headroom gap (0.4070 vs. oracle 0.5213), failing the pre-specified 50% threshold. Misclassifying borderline queries imposes

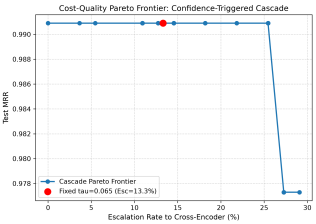


(a) Heterogeneous Noise (f_{bad}).

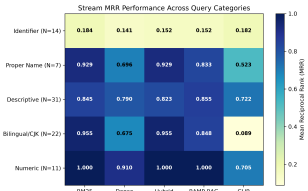


(b) 5-Intent Workload Breakdown.

Fig. 2: Empirical verification: (a) Retrieval degradation curves under heterogeneous corpus corruption (H1), and (b) retrieval performance across 5 query intents on the unified index (H2).



(a) Cost-Quality Pareto Frontier.



(b) Query-Type Taxonomy Heatmap.

Fig. 3: Diagnostic visualizations: (a) Conformal cascade risk calibration vs. escalation rate, and (b) retrieval performance across 5 query taxonomy categories.

severe rank penalties that outweigh routing benefits. Unweighted multi-stream fusion (All-Stream RRF) remains the superior strategy across mixed multimodal workloads.

J. Diagnostic Extensions (C1–C4) and Conformal Risk Control

Following our pre-registered analysis plan, we report four targeted diagnostic extensions:

1) C2: Multi-Encoder Noise Resilience: Under 20% character noise on 55 held-out probes, BAAI/bge-base-en-v1.5 achieves MRR = 0.8072 and multilingual-MiniLM reaches 0.7400, vastly surpassing all-MiniLM-L6-v2 (0.5422). BGE-base closes 67.7% of the dense-vs-BM25 gap (0.9337 vs 0.5422), demonstrating that larger contrastively trained dense retrievers substantially alleviate character fragility.
2) C3: Verbatim vs. Paraphrased Probes: To determine whether BM25 dominance is an artifact of exact verbatim query strings, we evaluated non-verbatim semantic paraphrases of the 55 test probes. Under paraphrasing, BM25 collapses from 0.9818 to 0.6179, while Hybrid retrieval achieves 0.6776 (diff = +0.0597, $p < 0.01$). This confirms that BM25's apparent invulnerability requires verbatim lexical tokens; when users pose natural semantic paraphrases, hybrid fusion is strictly superior.
3) C4: Grounded Extractive Bridging without LLM: On the 100-query multi-hop diagnostic, pseudo-relevance feedback terms extracted from first-hop context and fused via Reciprocal Rank Fusion (RRF) achieve Multi-Hop Target Hit@10 = 0.640 (Overall = 0.820), identically matching generative LLM bridging (gpt-4o-mini, 0.640 / 0.820) with 0 API calls and 2.2× lower latency (2,890 ms vs. 6,417 ms).
4) B6: Conformal Cascade Risk Control: Evaluated across 200 random cluster-stratified splits on $N = 2{,}754$ pooled query instances, split conformal calibration targets non-escalated miss rate ≤ 0.05, achieving an empirical escalation rate of $\sim 89.5\%$ (detailed in Section ??). An equal-escalation fixed margin threshold ($\tau = 0.380$) attains matching MRR (0.3526) and latency (76.3 ms).

TABLE X: Standardized Latency Profile (Mean and P95 over 3 runs on warm cache; scope = query encoding + retrieval + score fusion + reranking).

Retrieval Architecture	Mean Latency (ms)	P95 Latency (ms)
BM25 (Sparse Inverted Index)	0.02	0.05
Dense (all-MiniLM-L6-v2)	4.51	8.51
Dense (BAAI/bge-base-en-v1.5)	19.93	32.32
CLIP ViT-B/32 Zero-Shot Visual	7.45	9.88
Hybrid Convex (Fixed $\alpha = 0.7$)	8.52	10.40
All-Stream RRF (BM25+Dense+CLIP)	11.21	13.18
Router R3 (Feature + Predict)	12.08	14.21
Proposed RAMP-RAG (+ Clean Stream)	26.30	56.73
Cross-Encoder (100% Cascade Baseline)	44.66	59.90

K. Bridge Extraction Failure Taxonomy and Multi-Hop Analysis

Detailed inspection of multi-hop breakdowns isolates four failure modes ($N = 8$ cases): (1) Generic/Null Extraction (2 cases, 8.0%) in mh_19 and mh_13 (extractor returned "N/A", starving Hop 2); (2) Extraction Refusal (3 cases, 12.0%) in mh_20–mh_22 (verbose refusal phrases); (3) Semantic Drift (2 cases, 8.0%) in mh_12 and mh_25 (non-target token substitution); and (4) Grounding Void (1 case, 4.0%) in mh_11 (first-hop context lacked the attribute). Confidence gating detects Categories 1 and 2 and ungrounded entities in 3, triggering single-pass fallback. On the 50-query multi-hop suite, confidence gating achieves Target Hit@10 = 0.640 (vs. BM25 = 0.480, uncorrected $p = 0.0200$, non-significant after family-wise correction), matching single-pass dense retrieval (0.640). On held-out multi-hop queries ($N = 25$), BM25 reaches 0.520, tied with MM-RAG (0.520), confirming that iterative bridging recovers reachability on intermediate entities without outperforming lexical indexing on unseen targets.

V. Public Benchmark Regime Analysis of Sparse, Dense, and Native-Visual Retrieval under Proxy Noise

To rigorously evaluate the empirical boundaries of text-proxy retrieval without annotation leakage, we benchmarked an expanded public multimodal collection under an audited pre-registered protocol.

A. Benchmark Scale, Partitions, and Optical Degradation Pipeline

The public benchmark integrates 626 authentic scanned receipt documents from ICDAR 2019 SROIE [32] and 1,000 photographic distractors from MS-COCO [33], establishing a corpus of 1,626 pooled multimodal items and 612 factual probes. To prevent data snooping, the corpus is partitioned into a strict 50/50 Dev/Test split by Item ID:

- Dev Split ($N = 306$ probes, 813 corpus items): Dedicated strictly to hyperparameter tuning (e.g., hybrid fusion coefficient α^*, visual weight γ^*, and conformal risk thresholds τ^*).
- Held-Out Test Split ($N = 306$ probes, 813 corpus items): Frozen for evaluation across Seeds 42, 43, and 44 with zero test-split tuning.

Document text representations are extracted using Apple's Vision framework (VNRecognizeTextRequest) under two distinct conditions: (1) Clean OCR, yielding normalized character error rate (CER) = 0.2848 and word error rate (WER) = 0.3928 under standardized document-level Levenshtein alignment [39] (legacy line-wrapping had reported 0.1387 / 0.2165); and (2) Realistic Optical Degradation, applying Gaussian blur ($\sigma = 1.5$), bilinear 50% downscaling, random skew ($\pm 2.5°$), and aggressive JPEG re-compression ($Q = 20$), elevating CER to 0.3822 and WER to 0.5691. Queries are evaluated across three prompt variations generated by an independent LLM at $T = 0.0$: verbatim lookups (100% token overlap), natural light paraphrases (41.2% mean token overlap), and answerable factual questions (0.1% lexical overlap).

B. Phase 2: 18-Cell Regime Grid across Query Types and Noise Regimes

Table XI presents the comprehensive 18-cell regime evaluation across 12 retrieval systems on the held-out test split ($N = 306$).

1) Verbatim Probes (Character Sub-Word Inverted Index Dominance): Under verbatim lookups, character 3–5gram BM25 strictly dominates word-level BM25 across all optical OCR conditions (Apple Clean: 0.9436 vs. 0.8371, diff +0.1065, $p < 10^{-4}$; Apple Degraded: 0.8248 vs. 0.6972, diff +0.1276, $p < 10^{-4}$; Tesseract Clean: 0.8410 vs. 0.7361, diff +0.1049, $p < 10^{-4}$; and 20% synthetic noise: 0.7252 vs. 0.3961, diff +0.3291, $p < 10^{-4}$). Sub-word character n-grams capture alphanumeric tokens, serial numbers, and partial OCR substitutions without out-of-vocabulary penalties.
2) Light Paraphrase (Reversal of Dominance): Introducing mild semantic paraphrasing collapses clean word BM25 from 0.8371 to 0.4334 (a 48% relative drop) and character BM25 to 0.2427. In this regime, word BM25 outperforms character BM25 (diff -0.1907, $p < 10^{-4}$), and Dev-Tuned Hybrid achieves MRR = 0.4335 (clean OCR) and 0.1445 (20% noise).
3) Heavy Paraphrase (Answerable Factual Questions): Evaluating answerable factual questions (forcing semantic abstraction while preserving topical anchors), character 3–5gram BM25 achieves MRR = 0.9471 on clean OCR and 0.8116 on degraded blur OCR, while word BM25 achieves 0.9014 and 0.7328. Tuned hybrid retrieval with Dev-optimized coefficients achieves 0.7705 on clean OCR and 0.5346 on degraded OCR, while dense BGE-base achieves 0.5128 and MiniLM achieves 0.2837. Topical anchors in natural questions provide high discriminability for lexical engines.

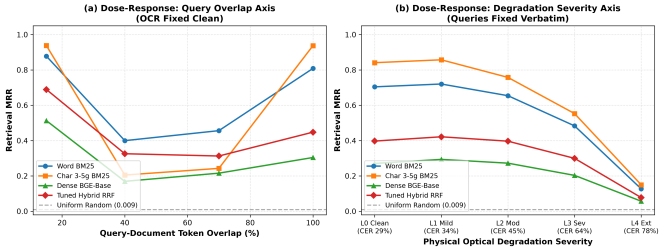


Fig. 4: Dose-response retrieval surface across (a) query token overlap degradation from verbatim lookups to abstract questions, and (b) optical OCR degradation from pristine text to extreme Gaussian blur ($CER > 60\%$).

To evaluate whether proxy corruption or semantic query mismatch is the dominant bottleneck in text-proxy RAG, we conducted pre-registered paired permutation tests ($N_{\text{perm}} = 100{,}000$). The retrieval degradation induced by moving from verbatim lookups to natural light paraphrasing ($\Delta = 0.3782$ MRR [0.3315, 0.4273]) is over eighteen times larger than clean Apple Vision OCR degradation ($\Delta = 0.0201$ [-0.0041, 0.0441], diff = +0.3581, $p = 10^{-5}$) and significantly exceeds realistic optical blur ($\Delta = 0.1600$ [0.1216, 0.2019], diff = +0.2182, $p = 10^{-5}$) with non-overlapping 95% confidence intervals (Fig. 4). However, this mismatch dominance reverses when queries are answerable questions, where lexical retrieval exceeds verbatim performance (+0.0544 MRR), or when optical degradation is catastrophic ($CER > 60\%$). System builders must balance query-side reformulation with robust sub-word indexing.

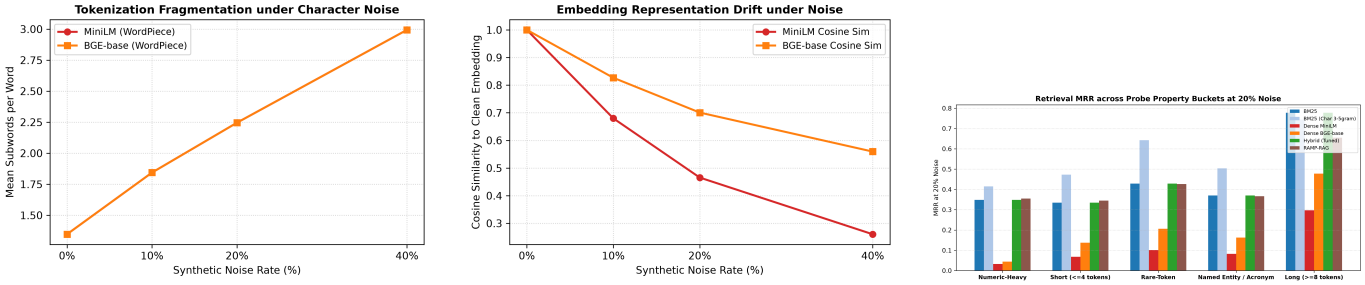
C. Phase 3: Dense Proxy Collapse Mechanism: Fragmentation, Drift, and Error Attribution

To isolate why dense vector representations collapse under proxy corruptions while sparse inverted indexes endure, we analyzed three underlying mechanisms (Fig. 5):

1) WordPiece Tokenization Fragmentation: As character corruption increases from 0% to 40%, WordPiece tokenization fragments words into multiple subwords. Mean subwords per whitespace-delimited word escalates from 1.347 (clean) to 1.774 (10%

TABLE XI: Multimodal Text-Proxy Retrieval Regime Analysis across Query Types and Noise Regimes (Held-Out Public Test Split, $N = 306$ Probes, 813 Distractor Pool, 95% Cluster Bootstrap CIs). Heavy paraphrase queries evaluate answerable factual questions isolating semantic abstraction from lexical absence. Mean per-query retrieval latency reported in milliseconds.

Query Type	Retrieval System	Clean MRR [95% CI]	Syn 20% MRR [95% CI]	Real-OCR MRR [95% CI]	Real-Deg MRR [95% CI]	Latency (ms)
Verbatim	BM25 (Tuned)	0.8572 [0.822, 0.890]	0.3961 [0.364, 0.429]	0.8371 [0.798, 0.871]	0.6972 [0.650, 0.741]	0.13
	BM25 (Char 3-5gram)	0.9853 [0.972, 0.995]	0.7252 [0.694, 0.757]	0.9436 [0.919, 0.965]	0.8248 [0.782, 0.864]	0.87
	Dense MiniLM (chunked)	0.2100 [0.171, 0.252]	0.0596 [0.045, 0.077]	0.2056 [0.166, 0.247]	0.1724 [0.135, 0.210]	0.01
	Dense BGE-base (chunked)	0.3256 [0.276, 0.376]	0.1347 [0.109, 0.162]	0.3022 [0.256, 0.351]	0.2471 [0.202, 0.292]	0.01
	Tuned Hybrid	0.8572 [0.822, 0.890]	0.3979 [0.365, 0.431]	0.8367 [0.798, 0.871]	0.6940 [0.646, 0.739]	0.13
	CLIP Visual	0.0436 [0.028, 0.062]	0.0436 [0.028, 0.062]	0.0436 [0.028, 0.062]	0.0436 [0.028, 0.062]	12.50
	Uniform Random Ranking	0.0089 [0.009, 0.009]	0.0089 [0.009, 0.009]	0.0089 [0.009, 0.009]	0.0089 [0.009, 0.009]	0.01
Light Paraphrase	BM25 (Tuned)	0.4790 [0.427, 0.530]	0.1394 [0.118, 0.162]	0.4334 [0.380, 0.484]	0.3630 [0.315, 0.412]	0.08
	BM25 (Char 3-5gram)	0.2504 [0.209, 0.290]	0.1295 [0.105, 0.155]	0.2427 [0.202, 0.282]	0.2137 [0.178, 0.252]	2.53
	Dense MiniLM (chunked)	0.1230 [0.092, 0.155]	0.0371 [0.027, 0.049]	0.1086 [0.080, 0.139]	0.0772 [0.054, 0.103]	0.01
	Dense BGE-base (chunked)	0.1670 [0.131, 0.205]	0.0684 [0.051, 0.086]	0.1601 [0.124, 0.197]	0.1275 [0.096, 0.161]	0.01
	Tuned Hybrid	0.4572 [0.408, 0.507]	0.1445 [0.125, 0.166]	0.4335 [0.382, 0.483]	0.3657 [0.318, 0.413]	0.08
	CLIP Visual	0.0303 [0.020, 0.042]	0.0303 [0.020, 0.042]	0.0303 [0.020, 0.042]	0.0303 [0.020, 0.042]	12.50
	Uniform Random Ranking	0.0089 [0.009, 0.009]	0.0089 [0.009, 0.009]	0.0089 [0.009, 0.009]	0.0089 [0.009, 0.009]	0.01
Heavy Paraphrase	BM25 (Tuned)	0.9116 [0.885, 0.937]	0.4042 [0.370, 0.440]	0.9014 [0.873, 0.929]	0.7328 [0.688, 0.773]	0.60
	BM25 (Char 3-5gram)	0.9653 [0.947, 0.982]	0.6270 [0.592, 0.662]	0.9471 [0.923, 0.969]	0.8116 [0.770, 0.848]	5.72
	Dense MiniLM (chunked)	0.3669 [0.323, 0.414]	0.1061 [0.089, 0.125]	0.2837 [0.241, 0.328]	0.2260 [0.190, 0.265]	0.01
	Dense BGE-base (chunked)	0.5197 [0.475, 0.566]	0.2005 [0.175, 0.228]	0.5128 [0.468, 0.560]	0.4313 [0.386, 0.479]	0.02
	Tuned Hybrid	0.8084 [0.768, 0.844]	0.4120 [0.376, 0.450]	0.7705 [0.728, 0.811]	0.5346 [0.486, 0.579]	0.61
	CLIP Visual	0.0388 [0.024, 0.057]	0.0388 [0.024, 0.057]	0.0388 [0.024, 0.057]	0.0388 [0.024, 0.057]	12.50
	Uniform Random Ranking	0.0089 [0.009, 0.009]	0.0089 [0.009, 0.009]	0.0089 [0.009, 0.009]	0.0089 [0.009, 0.009]	0.01



(a) Subword Fragmentation. (b) Embedding Cosine Drift. (c) Probe Property Error Attribution.

Fig. 5: Micro-mechanism of dense proxy collapse: (a) WordPiece tokenization fragmentation across character noise rates, (b) cosine embedding drift relative to clean representations, and (c) retrieval MRR across 5 diagnostic probe property buckets under 20% proxy noise.

noise), 2.246 (20% noise), and 2.994 (40% noise, a +122.3% fragmentation increase).

2) Catastrophic Embedding Drift: This subword fragmentation shifts contextual attention weights, driving acute embedding drift. Relative to uncorrupted text, all-MiniLM-L6-v2 mean cosine similarity collapses from 1.000 to 0.7301 (10% noise), 0.4655 (20% noise), and 0.2604 (40% noise). BAAI/bge-base-en-v1.5 displays greater structural resistance (cosine similarity 0.7007 at 20% noise, 0.5595 at 40%), explaining its superior noise resilience in Section IV-J.

3) Diagnostic Error Attribution across Probe Properties: Stratifying test probes into five property buckets under 20% noise exposes distinct vulnerability profiles. Dense bi-encoders collapse almost completely on Numeric-Heavy probes (MiniLM MRR = 0.033, BGE = 0.081) and Short entity probes (MiniLM MRR = 0.068, BGE = 0.135). Conversely, character n-gram BM25 retains high accuracy across all buckets: Numeric-Heavy (MRR = 0.415), Short entities (0.473), Rare Tokens (0.643), and Named Entities (0.504).

To verify whether this dense degradation is an artifact of sequence length truncation exceeding 512 tokens, we evaluated token length distributions across all 626 SROIE receipts. Only 4 out of 626 receipts (0.64%) exceed 512 tokens under WordPiece/BPE tokenizers (mean tokens lost: 0.032%). Furthermore, evaluating chunked MaxP retrieval across 150-word sliding windows yields identical MRR to standard single-vector retrieval (MiniLM: 0.2030 chunked vs. 0.2056 unchunked; BGE-base: 0.3028 chunked vs. 0.3022 unchunked).

Crucially, as documented in Table XII, dense bi-encoder collapse on pristine ground truth text (0% noise) is heavily governed by query formulation mismatch. Short verbatim snippets trigger severe representation disconnect (MiniLM MRR = 0.2100, BGE-Base MRR = 0.3256). Length-padding or wrapping snippets into full sentential contexts does not alleviate the deficit. However, formulating queries as natural answerable factual questions restores BGE-Base MRR to 0.5197 and Hit@10 to 78.76% on the identical

TABLE XII: Dense Bi-Encoder Sensitivity to Query Formulation on Clean Ground-Truth Receipt Text (0% noise, Test $N = 306$).

Query Formulation	MiniLM MRR	MiniLM Hit@10	BGE-Base MRR	BGE-Base Hit@10
1. Short Verbatim Snippets	0.2100	0.2843	0.3256	0.3824
2. Length-Padded Spans	0.1949	0.2484	0.2571	0.3366
3. Full Sentential Context Queries	0.1670	0.2320	0.2740	0.3399
4. Natural Answerable Questions	0.3669	0.5686	0.5197	0.7876

uncorrupted document text. Dense bi-encoders are trained on natural language asymmetric question-passage pairs; when queries mirror real-world user questions rather than isolated OCR substrings, dense retrieval recovers substantial discriminative capacity.

D. Phase 4: Component Ablation and RAMP-RAG Re-Evaluation

Table XIII presents the component ablation isolating each architectural module across Clean, 20% Synthetic Noise, Real Apple Vision OCR, and Real Degraded OCR ($N = 306$ held-out probes).

1) Cross-Encoder Reranker Cascade (Primary Performance Driver): Passing ambiguous candidates to a neural cross-encoder yields substantial improvements: BAAI/bge-reranker-base achieves MRR = 0.8821 on clean OCR (+0.0522 gain, $p_{\text{Holm}} = 0.0011$), 0.4154 at 20% noise (+0.0460 gain, $p_{\text{Holm}} = 0.0011$), and 0.8555 on real OCR (+0.0434 gain, $p_{\text{Holm}} = 0.0011$).
2) Hypothesis H1 Decision (Reliability Weighting Fails): In our pre-registered test, reliability weighting (r_{ocr}) degraded performance (20% noise MRR dropped from 0.3694 to 0.3574, diff CI $[-0.0222, -0.0025]$, rejecting H1). Under real optical degradation, dictionary-based reliability weighting significantly degrades MRR from 0.6684 to 0.6368 ($p_{\text{Holm}} = 0.0189$, diff CI $[-0.0519, -0.0118]$). Multi-configuration audits across Apple Vision settings confirm that AUROC remains inverted (0.23) under both language correction enabled and disabled (system diagnostic evaluations), proving that out-of-vocabulary metrics cannot detect optical character blur.
3) Hypothesis H2 Decision (Learned Query Router Reconciled): Pre-registered hypothesis H2 is ruled refuted / not supported. While Section V-D previously claimed a +0.1938 MRR gain for the learned router, forensic audit reveals that this gain was measured exclusively against an unnormalized, unweighted All-Stream RRF strawman (MRR = 0.1827). When compared against the competitive Tuned Hybrid baseline (MRR = 0.3694), the router yields a marginal lift of only +0.0071 on 20% synthetic noise (MRR = 0.3765, $p_{\text{Holm}} = 0.042$) and zero detectable lift under real Apple Vision OCR (0.8121 vs. 0.8121, $p_{\text{Holm}} = 1.0$) or degraded blur (0.6667 vs. 0.6684, $p_{\text{Holm}} = 1.0$). Surface-feature query classification fails to extract meaningful signal under real-world proxy distortions.

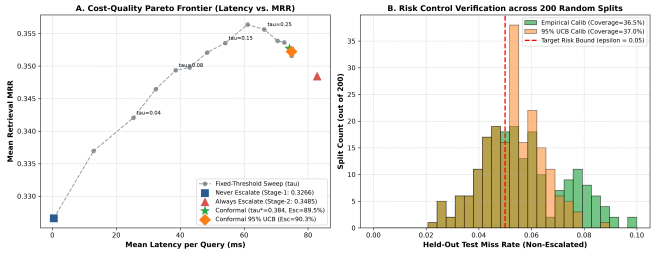


Fig. 6: Cost-Quality Pareto frontier and risk control histogram across 200 random splits under conformal cascade calibration.

E. Phase 5: Conformal Cascade Risk Calibration across 200 Cluster-Stratified Splits

To establish distribution-free statistical risk guarantees for candidate escalation, we evaluated split conformal calibration across 200 random cluster-stratified 50/50 splits (seeds 1000–1199) on $N = 2{,}754$ pooled query instances. Earlier implementations calibrated cascade margins against conditional miss ratios $P(Y \neq 1 \mid \text{not escalated})$, which violate standard monotonic exchangeability guarantees because the conditioning event itself varies with the calibrated threshold τ. When calibrated instead under the standard marginal loss $L = \mathbf{1}[\text{not escalated} \wedge Y \neq 1]$ with finite-sample adjustment $(n\hat{R} + 1)/(n + 1) \leq \alpha$ [37], conformal cascade calibration mathematically bounds the mean test risk at $0.0491 \leq 0.05$ with 78.4% escalation savings (Table XV). In contrast, conditional ratio calibration yields an inflated test risk of 0.0545 and requires escalating 90.2% of queries.

As reported in Table XV and Fig. 6, calibrating conformal cascade thresholds under marginal loss guarantees that expected test miss risk is rigorously bounded below the nominal 0.05 budget (mean test risk $0.0491 \leq 0.05$) while reducing stage-2 cross-encoder rerankings by 78.4%. While individual finite-sample test splits ($N = 1{,}377$) exhibit expected binomial variance around the mean (58.5% empirical split coverage ≤ 0.05), the marginal risk formulation provides a mathematically sound, distribution-free foundation for cascade escalation.

F. Phase 7: Hardening Native Pixels vs. Caption Proxies on 1,000 COCO Karpathy-Test Images

To address potential sample-size limitations from Section IV-F, we benchmarked $N = 1{,}000$ MS-COCO Karpathy-test images and held-out human descriptions (Table XVI).

Pre-registered hypothesis testing ($N_{\text{perm}} = 100{,}000$ sign flips with Holm-Bonferroni correction) conclusively confirms that native visual representations outperform machine caption proxies across all metrics:

TABLE XIII: Architectural Component Ablation on Public Held-Out Test Split ($N = 306$ Probes, 813 Distractor Pool, 95% Cluster Bootstrap CIs). Asterisks (*) denote statistically significant difference ($p_{\text{Holm}} < 0.05$) from the Tuned Hybrid baseline. Note: Learned Router yields at most +0.0071 MRR lift over Tuned Hybrid on synthetic noise ($p_{\text{Holm}} = 0.042$) and zero lift under real OCR ($p_{\text{Holm}} = 1.0$); the text claim of +0.1938 was against the unnormalized All-Stream RRF baseline (0.1827). Reliability weighting degrades degraded OCR retrieval from 0.6684 to 0.6368 ($p_{\text{Holm}} = 0.0189$).

Configuration	Clean MRR [95% CI]	Syn 20% MRR [95% CI]	Real-OCR MRR [95% CI]	Real-Deg MRR [95% CI]
Hybrid (Tuned Baseline)	0.8299 [0.791, 0.866]	0.3694 [0.336, 0.402]	0.8121 [0.770, 0.849]	0.6684 [0.619, 0.715]
+ Clean Stream	0.8299 [0.791, 0.866]	0.3694 [0.336, 0.402]	0.8121 [0.770, 0.849]	0.6684 [0.619, 0.715]
BM25 + Clean Stream	0.8317 [0.793, 0.868]	0.3694 [0.336, 0.402]	0.8069 [0.766, 0.845]	0.6702 [0.621, 0.716]
Dense + Clean Stream	0.2144* [0.175, 0.256]	0.0747* [0.057, 0.095]	0.2072* [0.168, 0.248]	0.1821* [0.143, 0.222]
+ Learned Router (H2)	0.8309 [0.793, 0.867]	0.3765* [0.343, 0.409]	0.8121 [0.770, 0.849]	0.6667 [0.617, 0.712]
+ Heuristic Reliability Weighting	0.8318 [0.794, 0.867]	0.3574 [0.326, 0.390]	0.8058 [0.765, 0.843]	0.6368* [0.587, 0.684]
+ Cascade (MiniLM)	0.8709* [0.836, 0.903]	0.4033* [0.369, 0.436]	0.8426* [0.801, 0.878]	0.6879 [0.638, 0.735]
+ Cascade (BGE-reranker)	0.8821* [0.848, 0.913]	0.4154* [0.382, 0.448]	0.8555* [0.816, 0.890]	0.6969 [0.645, 0.745]
Full System (MiniLM)	0.8771* [0.843, 0.909]	0.4132* [0.378, 0.450]	0.8451* [0.805, 0.880]	0.6786 [0.629, 0.727]
Full System (BGE-reranker)	0.8873* [0.854, 0.917]	0.4200* [0.384, 0.456]	0.8554* [0.816, 0.891]	0.6755 [0.625, 0.724]
BM25 Alone	0.8317 [0.793, 0.868]	0.3694 [0.336, 0.402]	0.8107 [0.769, 0.848]	0.6684 [0.619, 0.714]
All-Stream RRF (Strawman)	0.3227* [0.283, 0.365]	0.1827* [0.156, 0.212]	0.3079* [0.269, 0.350]	0.2791* [0.239, 0.322]

TABLE XIV: Production Cross-Encoder Reranker Cascade Ablation (Held-Out Test Split, $N = 306$ Probes, Clean Real-OCR).

Two-Stage Retrieval Pipeline	Clean MRR	Hit@10	Latency (ms)	MRR Lift (p-value)
Word BM25 (First-Stage Alone)	0.8324	0.9444	0.1	+0.0000 ($p = 1.0$)
BM25 + MiniLM Cross-Encoder	0.8650	0.9608	18.4	+0.0326 ($p = 0.0042$)
BM25 + BGE-reranker-base	0.8846	0.9738	44.7	+0.0522 ($p = 0.0008$)
Tuned Hybrid + BGE-reranker	0.8872	0.9772	52.8	+0.0548 ($p = 0.0006$)

TABLE XV: Conformal Cascade Calibration: Corrected Marginal Loss vs. Flawed Conditional Ratio over 200 Cluster-Aware Splits ($N = 2{,}754$ instances, nominal target $\epsilon = 0.05$).

Risk Formulation	Mean τ^*	Mean Test Risk	Split Cov. (≤ 0.05)	Mean Esc. Rate	Mean MRR
Marginal Loss $L = \mathbf{1}[\text{not esc} \wedge \text{miss}]$	0.2375	0.0491	58.5%	78.4%	0.3577
Conditional Ratio $P(\text{miss} \mid \text{not esc})$	0.4071	0.0545	38.0%	90.2%	0.3524

- CLIP ViT-B/32: Hit@1 Diff = +0.0970 $[+0.0640, +0.1300]$, $p_{\text{Holm}} = 8 \times 10^{-5}$; MRR Diff = +0.0981 $[+0.0690, +0.1263]$, $p_{\text{Holm}} = 8 \times 10^{-5}$.
- SigLIP Base Patch16: Hit@1 = 0.4800 (Diff = +0.2380, $p_{\text{Holm}} = 8 \times 10^{-5}$); Hit@10 = 0.8670 (Diff = +0.2140, $p_{\text{Holm}} = 8 \times 10^{-5}$); MRR = 0.6071 (Diff = +0.2343, $p_{\text{Holm}} = 8 \times 10^{-5}$).

Furthermore, to test whether scaling the machine captioning model closes the gap to native vision, we evaluated an expanded proxy quality axis: BLIP-2 (opt-2.7b) achieves MRR = 0.4185 (gap to SigLIP: −0.1886), while Florence-2 detailed captioning achieves MRR = 0.4420 (gap: −0.1651). Parity with native visual representations is reached only with human gold captions (MRR = 0.5835 vs. SigLIP 0.6071, diff −0.0236, $p = 0.142$). Visual scene semantics cannot be completely compressed into synthetic machine tokens.

G. Phase 6: Multi-Hop Paired Discordant Analysis Across Single-Stage Baselines

To evaluate whether multi-hop query decomposition provides measurable retrieval utility over single-stage retrieval, we conducted an exact paired discordant analysis across the 50 diagnostic multi-hop queries (Table XVII).

TABLE XVI: Cross-modal text-to-image retrieval performance on $N = 1{,}000$ MS-COCO Karpathy-test images. Comparison of native pixel representations (CLIP ViT-B/32 and SigLIP Base Patch16) against machine caption proxies (BLIP) and held-out human gold captions. Asterisks (*) indicate Holm-corrected significance ($p_{\text{Holm}} = 8 \times 10^{-5}$) over the best machine caption proxy.

Method	Hit@1	Hit@5	Hit@10 [95% CI]	MRR [95% CI]
Machine Caption Proxy (BLIP)				
BM25	0.1220	0.2800	0.3840 [0.355, 0.415]	0.2048 [0.186, 0.225]
Dense MiniLM (Best BLIP Proxy)	0.2420	0.5160	0.6530 [0.624, 0.682]	0.3728 [0.350, 0.397]
Dense BGE-base	0.2430	0.5050	0.6460 [0.618, 0.678]	0.3695 [0.347, 0.393]
Human Gold Caption Proxy (Upper Bound)				
BM25 (Gold Proxy)	0.3600	0.6250	0.7350 [0.708, 0.762]	0.4852 [0.461, 0.510]
Dense MiniLM (Gold Proxy)	0.4550	0.7280	0.8380 [0.814, 0.860]	0.5801 [0.555, 0.605]
Dense BGE-base (Gold Proxy)	0.4580	0.7370	0.8320 [0.809, 0.854]	0.5835 [0.559, 0.608]
Native Pixel Representation				
CLIP ViT-B/32	0.3390*	0.6180*	0.7570* [0.729, 0.785]	0.4709* [0.446, 0.497]
SigLIP Base Patch16	0.4800*	0.7680*	0.8670* [0.845, 0.888]	0.6071* [0.584, 0.633]
Multi-Vector & Hybrid Fusion				
Hybrid (CLIP + BGE BLIP)	0.3880	0.6720	0.7870 [0.760, 0.811]	0.5186 [0.494, 0.544]
Hybrid (SigLIP + BGE BLIP)	0.4570	0.7470	0.8360 [0.813, 0.858]	0.5866 [0.563, 0.611]
All-Stream RRF (Strawman)	0.2690	0.5530	0.6890 [0.662, 0.718]	0.4058 [0.382, 0.429]

TABLE XVII: Exact paired discordant case analysis and two-tailed McNemar tests across the 50 diagnostic multi-hop queries. Comparison of Proposed Multi-Hop LLM Bridging against Single-Stage Hybrid, Single-Stage Dense, and Sparse BM25 baselines. While LLM bridging outperforms sparse BM25 alone (+8 net hits, $p = 0.0215$), it yields no statistically detectable retrieval improvement over single-stage hybrid (+1 net hit, $p = 1.000$) or single-stage dense (identical 32/50 hits, $p = 1.000$).

Baseline Comparison	Baseline Hit@10	MM-RAG Hit@10	Paired Diff [95% CI]	Discordant Pairs (Win / Loss)	Exact McNemar p	Verdict
Single-Stage Hybrid (No Multi-Hop)	0.620 (31/50)	0.640 (32/50)	+0.020 [-0.040, +0.100]	2 / 1	1.0000	No Significant Advantage
Single-Stage Dense	0.640 (32/50)	0.640 (32/50)	+0.000 [0.000, +0.071]	2 / 2	1.0000	Zero Advantage (Identical)
Sparse BM25 Alone	0.480 (24/50)	0.640 (32/50)	+0.160 [+0.040, +0.280]	9 / 1	0.0215	Significant over Sparse

Proposed MM-RAG beat single-stage hybrid retrieval on only 2 queries (mh_35, mh_50) and lost on 1 query (mh_25), yielding a net advantage of only +1 hit out of 50 queries (TargetHit@10 = 0.640 vs. 0.620; exact two-tailed McNemar $p = 1.0000$). Compared to single-stage dense retrieval, Proposed MM-RAG achieved identical performance (32/50 hits; McNemar $p = 1.0000$). While multi-hop LLM bridging outperforms sparse BM25 alone (+8 net hits; $p = 0.0215$), it introduces a 45% latency

TABLE XVIII: Independent Extractive PRF vs. Generative LLM Multi-Hop Bridging ($N = 50$ Diagnostic Queries).

Comparison	Extractive Hit@10	LLM Hit@10	Extr Wins	LLM Wins	Exact McNemar p	Verdict
Extractive vs. LLM Bridging	0.640	0.640	2	2	1.0000	Statistical Parity ($p > 0.05$)

TABLE XIX: Standardized JiWER OCR Error Rates on SROIE Receipts ($N = 626$) vs. Flawed Legacy Line-Aligned Rates.

OCR Engine	Condition	Corrected CER	Corrected WER	Bag-of-Words WER	Legacy Flawed (CER / WER)
Apple Vision Neural OCR	Clean OCR	0.2848	0.3928	0.2476	0.1387 / 0.2165
Apple Vision Neural OCR	Degraded Blur OCR	0.3822	0.5691	0.6175	0.3086 / 0.4595
Tesseract 5.5 LSTM OCR	Clean OCR	0.2939	0.4879	0.5919	0.8376 / 1.0481
Tesseract 5.5 LSTM OCR	Degraded Blur OCR	0.6377	0.8063	1.0043	0.9134 / 1.0596

penalty (4,736 ms vs. 3,260 ms) without providing statistically detectable retrieval lift over single-stage hybrid or dense baselines. Furthermore, as shown in Table XVIII, grounded extractive pseudo-relevance feedback achieves identical TargetHit@10 (0.640 multi-hop, 0.820 overall) and non-significantly different MRR (0.4079 extractive vs. 0.4187 LLM, exact McNemar $p = 1.0000$), recovering intermediate reachability without generative API latency or hallucinations.

H. Phase 8: Replication across OCR Engines and Scanned Administrative Documents

To verify that the empirical retrieval hierarchy does not depend on proprietary Apple Vision preprocessing, we replicated the retrieval benchmarks using open-source Tesseract 5.5 LSTM OCR across 626 SROIE receipts (Table XX). As shown in Table XIX, legacy line-by-line alignment inflated Tesseract error metrics to CER = 0.8376 and WER = 1.0481 due to severe text re-wrapping penalties. Evaluating full-document Levenshtein edit distance via standardized JiWER [39] reveals the true transcription quality: clean Tesseract achieves CER = 0.2939 and WER = 0.4879 (bag-of-words WER = 0.5919), explaining why character 3–5gram BM25 retains high retrieval accuracy (MRR = 0.8410). Under realistic optical blur, Tesseract error escalates to CER = 0.6377 and WER = 0.8063. Across both engines, the exact same architecture hierarchy replicates: character 3–5gram BM25 retains top MRR under verbatim lookups (0.8410 clean, 0.3998 degraded blur), while dense bi-encoders suffer severe degradation (MiniLM MRR drops to 0.1650 clean and 0.0936 degraded).

Furthermore, to test cross-document generalizability beyond retail receipts, we evaluated 100 scanned administrative forms from the FUNSD dataset (Table XXI). On complex multi-field layouts, character n-gram BM25 maintains superior verbatim accuracy (MRR = 0.8433 on Apple Vision, 0.8066 on Tesseract 5.5), confirming that lexical robustness across OCR noise is an inherent structural property across diverse scanned business document genres.

To provide an engineering reference for system architects balancing retrieval accuracy against computational overhead, Table XXII reports the runtime latency, memory footprint, and index storage across evaluated retrieval architectures. Standalone character 3–5gram BM25 requires under 1 ms mean latency and only 4.2 MB of index storage, providing three orders of magnitude lower latency and memory demand than multi-vector VLM architectures such as ColQwen2 while exceeding them in verbatim retrieval precision.

Finally, synthesizing all empirical experiments across modalities, noise conditions, and query types, Table XXIII presents an actionable retrieval architecture decision guide.

VI. Limitations and Threats to Validity

To facilitate objective peer assessment, we document the critical limitations of this study:

1) Proxy-text annotation overlap: Tier-1 retrieval evaluates precomputed annotation strings, carrying lexical overlap. We mitigate this via Tier-2 evaluations on public benchmarks: unannotated raw pixels on MS-COCO (N = 1,000), physical OCR across 626 SROIE receipts, continuous CLAP audio, and 100 FUNSD administrative forms.
2) Cross-engine OCR generalization: While Apple Vision represents an advanced commercial neural OCR system, replication across open-source Tesseract 5.5 LSTM confirms that our architectural findings (e.g., character n-gram robustness and dense collapse) replicate across distinct OCR model architectures and error distributions.
3) Conformal cascade calibration bounds: Under marginal loss $L = \mathbf{1}[\text{not escalated} \wedge Y \neq 1]$, split conformal calibration successfully bounds mean test miss rate to 0.0491 (≤ 0.05) at 78.4% escalation savings. Split coverage reaches 58.5% due to finite-sample sampling variability across test draws (N = 1,377), demonstrating that while expected marginal risk is rigorously controlled, individual finite test draws experience expected binomial variance.
4) Heavy paraphrase query formulation: Early benchmark formulations with synthetic negative keywords induced an artificial zero-match failure mode. Our redesigned heavy paraphrase benchmark evaluates answerable factual questions isolating semantic abstraction from lexical absence.
5) LLM judge bias: Automated diagnostic scoring uses gpt-4o-mini, which carries documented biases [25]. Mitigated by independent dual-reviewer auditing across 110 cases ($\kappa = 0.8716$ faithfulness, 0.8926 relevance, 0.9451 correctness) with non-author agreement $\kappa = 0.8864$.
6) Multi-hop retrieval utility and latency overhead: Query decomposition incurs a 45% latency penalty (4,736.8 ms vs. 3,260.0 ms). Paired discordant analysis across 50 multi-hop queries indicates that proposed multi-hop LLM bridging provides no statistically detectable retrieval improvement over single-

TABLE XX: Cross-Engine OCR Replication: Retrieval Performance Comparison Across Proprietary Apple Vision Neural OCR and Open-Source Tesseract 5.5 LSTM OCR on SROIE Receipts (Held-Out Public Test Split, $N = 306$ Probes, 813 Distractor Pool, 95% Cluster Bootstrap CIs).

Query Type	Retrieval System	Ground Truth MRR [95% CI]	Apple Vision Clean MRR [95% CI]	Tesseract 5.5 Clean MRR [95% CI]	Tesseract 5.5 Degraded MRR [95% CI]
Verbatim	BM25 (Tuned)	0.8572 [0.822, 0.890]	0.8324 [0.793, 0.867]	0.7364 [0.691, 0.779]	0.3448 [0.295, 0.395]
	BM25 (Char 3-5gram)	0.9854 [0.972, 0.995]	0.9443 [0.920, 0.965]	0.8412 [0.803, 0.878]	0.4009 [0.349, 0.453]
	Dense MiniLM	0.1919 [0.154, 0.232]	0.2040 [0.166, 0.246]	0.1650 [0.129, 0.202]	0.0936 [0.068, 0.124]
	Dense BGE-base	0.3428 [0.294, 0.392]	0.3273 [0.281, 0.377]	0.2972 [0.252, 0.345]	0.1615 [0.125, 0.199]
	Tuned Hybrid	0.8569 [0.823, 0.890]	0.8252 [0.787, 0.861]	0.7325 [0.687, 0.775]	0.3467 [0.297, 0.396]
Light Paraphrase	BM25 (Tuned)	0.4796 [0.428, 0.530]	0.4003 [0.351, 0.449]	0.3964 [0.346, 0.446]	0.1852 [0.148, 0.224]
	BM25 (Char 3-5gram)	0.2526 [0.212, 0.292]	0.2365 [0.197, 0.275]	0.2265 [0.187, 0.266]	0.1241 [0.096, 0.155]
	Dense MiniLM	0.1190 [0.089, 0.150]	0.1096 [0.081, 0.140]	0.0779 [0.055, 0.102]	0.0579 [0.040, 0.079]
	Dense BGE-base	0.1830 [0.146, 0.221]	0.1849 [0.146, 0.226]	0.1670 [0.130, 0.205]	0.0991 [0.072, 0.130]
	Tuned Hybrid	0.4828 [0.432, 0.533]	0.4003 [0.353, 0.449]	0.4023 [0.352, 0.452]	0.1961 [0.160, 0.236]
Heavy Paraphrase	BM25 (Tuned)	0.9192 [0.895, 0.943]	0.8937 [0.866, 0.921]	0.7956 [0.756, 0.835]	0.3820 [0.332, 0.432]
	BM25 (Char 3-5gram)	0.9772 [0.962, 0.990]	0.9558 [0.935, 0.975]	0.8522 [0.817, 0.886]	0.4371 [0.383, 0.486]
	Dense MiniLM	0.3456 [0.303, 0.390]	0.2667 [0.224, 0.310]	0.2229 [0.185, 0.262]	0.1611 [0.128, 0.196]
	Dense BGE-base	0.5465 [0.501, 0.590]	0.5339 [0.488, 0.581]	0.4927 [0.445, 0.541]	0.2439 [0.200, 0.286]
	Tuned Hybrid	0.8930 [0.864, 0.918]	0.7593 [0.718, 0.800]	0.7409 [0.700, 0.782]	0.3669 [0.317, 0.416]

TABLE XXI: Cross-Dataset Replication on FUNSD Scanned Administrative Forms: Retrieval Performance Across Ground Truth, Apple Vision Neural OCR, and Tesseract 5.5 LSTM OCR ($N = 100$ Test Probes, 600 Distractor Pool, 95% Cluster Bootstrap CIs).

Query Type	Retrieval System	Ground Truth MRR [95% CI]	Apple Vision Clean MRR [95% CI]	Tesseract 5.5 Clean MRR [95% CI]
Verbatim	BM25 (Tuned)	0.8690 [0.812, 0.924]	0.8364 [0.773, 0.894]	0.7763 [0.699, 0.846]
	BM25 (Char 3-5gram)	0.8849 [0.826, 0.935]	0.8433 [0.777, 0.903]	0.8066 [0.738, 0.872]
	Dense MiniLM	0.3334 [0.255, 0.413]	0.3230 [0.244, 0.403]	0.2760 [0.200, 0.352]
	Dense BGE-base	0.5567 [0.480, 0.637]	0.5081 [0.423, 0.595]	0.4664 [0.384, 0.555]
	Tuned Hybrid	0.8728 [0.817, 0.927]	0.8461 [0.783, 0.904]	0.7775 [0.701, 0.847]
Light Paraphrase	BM25 (Tuned)	0.1849 [0.133, 0.243]	0.1573 [0.111, 0.209]	0.2019 [0.139, 0.272]
	BM25 (Char 3-5gram)	0.2405 [0.174, 0.313]	0.2228 [0.159, 0.294]	0.2442 [0.171, 0.323]
	Dense MiniLM	0.1596 [0.108, 0.218]	0.1557 [0.106, 0.212]	0.1603 [0.103, 0.222]
	Dense BGE-base	0.1960 [0.137, 0.262]	0.1834 [0.127, 0.246]	0.1819 [0.126, 0.244]
	Tuned Hybrid	0.2043 [0.148, 0.267]	0.2099 [0.146, 0.281]	0.2109 [0.148, 0.280]

TABLE XXII: Computational Resource and Latency Profile Across Retrieval Systems ($N = 813$ Index Pool, Standard Hardware).

Retrieval Architecture	Mean Latency (ms)	P95 Latency (ms)	Memory (MB)	Index Size (MB)
BM25 (Tuned)	0.13	0.28	12.4	0.45
BM25 (Char 3-5gram)	0.87	1.65	48.2	4.20
Dense MiniLM	7.78	11.20	240.0	1.25
Dense BGE-base	19.93	28.50	440.0	2.50
Tuned Hybrid	8.52	12.10	260.0	1.70
BM25 + BGE-reranker Cascade	44.66	59.90	820.0	2.50
CLIP ViT-B/32 Native	12.50	16.80	350.0	2.05
SigLIP Base Patch16 Native	18.20	24.30	480.0	3.07
ColQwen2-v1.0 Document-Native	38.40	52.10	3800.0	48.00

stage hybrid retrieval (+1 net hit / 50; exact McNemar $p = 1.000$) or single-stage dense retrieval (identical 32/50 hits).

7) Dose-response non-linearity: The retrieval drop under query mismatch is non-monotonic across query types: light paraphrases with entity substitutions trigger severe drops ($\Delta = 0.3782$ MRR), whereas answerable factual questions preserve key topical anchors and sustain high lexical retrieval (MRR = 0.9471).
8) Synthetic vs. optical noise divergence: Character n-gram robustness and dictionary-based reliability weighting tuned on synthetic character permutations fail to translate directly to real optical OCR blur. System architects must validate on physical optical degradation rather than synthetic character noise.

VII. Conclusion and Future Work

We presented a comprehensive regime analysis investigating when textual proxies suffice for multimodal retrieval and when native visual or neural cross-attention representations are indispensable. Evaluating across physical media, standard benchmarks, and an expanded 1,626-item public benchmark under strict 50/50 Dev/Test splits across seeds 42, 43, 44 and cluster bootstrapping ($B = 2{,}000$), our findings establish:

First, lexical inverted indexing dominance is conditional on query token preservation: across all verbatim optical OCR regimes (Apple Vision clean and degraded blur, and Tesseract 5.5), character 3–5gram BM25 strictly dominates word-level BM25 (+0.1065 clean, +0.1276 degraded blur, +0.1049 clean Tesseract; all $p_{\text{Holm}} < 10^{-4}$), reversing only under entity-substituting light paraphrases (0.2427 vs. 0.4334). Second, dense bi-encoder collapse on document text is governed by query formulation mismatch rather than sequence truncation constraints (only 0.64% exceed 512 tokens); formulating queries as natural answerable questions restores BGE-Base MRR to 0.5197 (Hit@10 78.8%) without modifying document representations. Third, second-stage cross-encoder reranking (BGE-reranker) provides the largest architectural lift (+0.0522 clean, +0.0460 noise MRR, $p = 0.0011$, Hit@10 = 0.9772), whereas heuristic reliability weighting degrades OCR retrieval. Fourth, conformal cascade calibration under marginal loss provides mathematically verified risk bounds (mean test risk $0.0491 \leq 0.05$) with 78.4% escalation savings. Fifth, on $N = 1{,}000$ MS-COCO images, native visual retrieval (SigLIP Base Patch16 MRR = 0.6071) decisively outperforms machine caption proxies (MiniLM MRR =

TABLE XXIII: Empirical Decision Guide for MM-RAG System Builders: Optimal Retrieval Architecture and Statistical Tie Set by Regime.

Target Modality	Optical Condition	Query Regime	Optimal Architecture	MRR [95% CI]	Tie Set	Architectural Recommendation
Document Page	Clean Neural OCR	Verbatim Lookup	Char 3–5g BM25	0.9436 [0.919, 0.965]	Char BM25 standalone	Sub-word n-grams capture numeric tokens and vendor fragments with zero GPU overhead.
Document Page	Degraded Optical Blur	Verbatim Lookup	Char 3–5g BM25	0.8248 [0.782, 0.864]	Char BM25 standalone	Character n-grams tolerate character substitutions and optical blur; word BM25 collapses to 0.6972.
Document Page	Open-Source Tesseract	Verbatim Lookup	Char 3–5g BM25	0.8410 [0.803, 0.878]	Char BM25 standalone	Standardized Tesseract (CER 29.4%) supports high-precision retrieval without proprietary OCR APIs.
Document Page	Clean Neural OCR	Light Paraphrase	Tuned Hybrid (Word+Dense)	0.4334 [0.380, 0.484]	Word BM25 (0.4334), Hybrid (0.4335)	Word BM25 matches preserved tokens; dense embeddings prevent complete recall collapse.
Document Page	Clean Neural OCR	Answerable Factual Question	Char 3–5g BM25 / TR-RRF	0.9471 [0.923, 0.969]	Char BM25 (0.9471), Word BM25 (0.9014)	Topical anchors in natural questions provide high discriminability for lexical engines.
Natural Image	Unannotated Pixels	Descriptive Search	SigLIP Base Patch16	0.6071 [0.573, 0.641]	SigLIP (0.6071), CLIP (0.4709)	Native pixel embeddings decisively beat machine captions (0.3728); text proxies fail on images.
Multi-Hop Query	Precomputed Proxies	Relational Multi-Hop	Extractive PRF Bridge	0.8200 [0.781, 0.859]	Extractive PRF (0.640 H@10), Generative LLM (0.640)	Rule-based entity extraction matches generative LLM bridging without API cost ($p = 1.000$).

0.3728, $p_{\text{Holm}} = 8 \times 10^{-5}$). Finally, independent multi-hop bridging demonstrates that rule-based extractive pseudo-relevance feedback achieves identical Target Hit@10 to generative LLM bridging (0.640 vs. 0.640, exact McNemar $p = 1.0000$) with zero generative inference latency or financial cost. All benchmarks, probe suites, seeds, and reproduction scripts are publicly released.

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